Lab 5 Report

Fluid Dynamics

Matthew Pepper

MAE 315 M003

Prof Matthew Erdman

Part A Data Collected: November 6, 2025

Part B Data Collected: November 20, 2025

Due: December 16, 2025

1 Abstract

This report presents six wind tunnel experiments investigating fundamental fluid dynamics principles including aerodynamic forces, flow characterization, and boundary layer behavior. Wind tunnel calibration established a quadratic velocity-frequency relationship ($v = 0.0022f^2 + 0.8761f - 0.5423$, $R^2 = 0.9993$) used throughout subsequent testing. Vortex shedding measurements on a circular cylinder yielded frequencies of 100.391 Hz and 151.172 Hz at 30 Hz and 45 Hz fan speeds, within 13-16% of theoretical predictions. Cylinder wake measurements produced drag coefficients of 0.924 and 0.670 at Reynolds numbers of 7.519e+4 and 1.396e+5, consistent with published data. Boundary layer characterization revealed velocity gradients demonstrating no-slip conditions and downstream thickening. Clark Y airfoil testing in clean and dirty configurations showed maximum lift-to-drag ratios of ~6.5 for clean and ~3.5 for dirty setups, with dirty configurations extending stall from 9° to 15° angle of attack. Cylinder length comparison demonstrated that the longest cylinder (2D) achieved the highest drag coefficient (0.2721) due to increased frontal area despite reduced three-dimensional effects. These experiments collectively demonstrate fundamental aerodynamic principles governing lift generation, drag production, flow separation, and boundary layer behavior critical to aircraft design, structural engineering, and fluid-structure interaction analysis.

Contents

1 Abstract (2)

2 Introduction (4)

3 Procedure (12)

4 Results (15)

4.1 Wind Tunnel Calibration (15)

4.2 Vortex Shedding (17)

4.3 Cylinder Wake (20)

4.4 Boundary Layer (23)

4.5 Airfoil (25)

4.6 2D/3D Cylinders (35)

5 Conclusion (37)

6 Appendix (38)

6.1 References (38)

6.2 Plots and Tables (39)

6.3 Code (56)

~~6.4 Images of Published Code (71)~~
Deleted for purpose of pdf size for internship application

6.5 AI Statement (~~104~~) (71)

2 Introduction

Fluid dynamics is essential to understanding how forces and motion arise in fluid flow, and wind-tunnel testing is a useful experimental method used in studying aerodynamics. The purpose of this lab was to characterize air behavior and aerodynamic loading using a combination of pressure based velocity measurements, wake characterization behind a cylinder, vortex shedding analysis, boundary-layer concepts, nondimensional coefficients (lift, drag, pitching moment coefficients), and aerodynamic force and moment measurements using 6-DOF force balance. Each of these experiments illustrates essential principles used in the design of aircraft, wind turbines, vehicles, and any system where fluid-structure interactions are important.

Wind tunnel testing is governed by the conservation of mass and momentum. For steady, incompressible flow, the continuity equation ensures mass conservation through any control volume:

$$\frac{\delta}{\delta t}\int_{C\forall} \rho \, d\forall + \int_{CS} \rho V \cdot n dA = 0$$

Momentum conservation provides a second essential relation. The integral momentum equation balances the rate of momentum change within a control volume and the momentum flux across its boundaries with the forces acting on the fluid:

$$\frac{\delta}{\delta t}\int_{C\forall} v\rho d\forall + \int_{CS} v\rho(v \cdot n) dA = \sum F$$

When the continuity and momentum equations are combined with viscous stresses, they yield the Navier-Stokes equations, which describe fluid motion at every point in space:

$$\frac{\delta v}{\delta t} + (v \cdot \nabla)v = gh - \frac{1}{\rho}\nabla P + \upsilon\nabla^2 v$$

A key non-dimensional parameter associated with these equations is the Reynolds number,

$$Re = \frac{\rho U D}{\mu}$$

which compares inertial and viscous forces. Its magnitude indicated whether its laminar or turbulent fluid flow. Lower values correspond to laminar flow, while higher values indicate turbulent behavior. For a circular cylinder, fully turbulent wake development typically occurs when Re ~ 3e+5.

At Mach<<1 used in our wind tunnel, where density changes are negligible, the flow can be treated as incompressible. For steady, low viscosity flow, the Navier-Stokes equations simplify to Bernoulli's equation:

$$P + \frac{1}{2}\rho U^2 + \rho g h = constant$$

If elevation differences are small, this relation identifies three important pressures, static pressure, P, dynamic pressure, $P_{dyn} = \frac{1}{2}\rho U^2$, and stagnation pressure.

The pitot-static tube used in the calibration portion of the lab measures stagnation and static pressure simultaneously. Their difference gives the dynamic pressure, which yields the freestream velocity:

$$U = \sqrt{\frac{2(P_0 - P)}{\rho}}.$$

Because the wind tunnel control system reports only the fan frequency (in Hz), the first part of the experiment involved developing a calibration curve that converts fan speed to airflow velocity using this pressure velocity relationship. This calibration is essential for interpreting later measurements such as wake, drag forces, and vortex shedding frequencies. We used MATLAB's polyfit function to create a line of best fit. The equation derived is below:

$$0.0022x^2 + 0.8761x - 0.5423$$

With an R^2 value of 0.9993, I can conclude my line of best fit is very accurate as it is very close to 1. We use this equation to extrapolate velocity values beyond what was recorded in our experiment. In our lab, we only took measurements up until 40 Hz, we used the derived equation to extrapolate accurate velocity for 40 and 45 Hz.

To analyze drag forces later in the lab, we applied control volume analysis to relate the flow entering and leaving the test section. A control volume defines a region in space used to track how pressure and velocity distributions influence the net force on an object. In our case, the inlet of the control volume corresponds to the upstream flow in the tunnel, while the outlet corresponds to the pressure and velocity distribution measured downstream of the model.

The starting point is the integral momentum equation applied to the control volume:

$$\sum F = \frac{\delta}{\delta t}\int_{CV} v\rho \, dV + \int_{CS} v\rho v \cdot n \, dA = -\int\int_{CS} P \, dA - F_0$$

Assume steady flow:

$$\frac{\delta}{\delta t} = 0$$

And:

$$-\int\int_{CS} P \, dA = 0$$

So we have:

$$\sum F = \int_{CS} v\rho v \cdot n\, dA = -F_0$$

And from conservation of mass for an inertial control volume:

$$\frac{\delta}{\delta t}\int_{CV} \rho\, dV + \int_{CS} \rho v \cdot n\, dA = 0$$

Momentum equation in x-direction:

$$-F_0 = \int_{CS} u\rho v \cdot n\, dA = \int_0^w u_1 \rho(-u_1)\, dy + \int_0^w u_2 \rho u_2\, dy$$

$$F_0 = \int_0^w \rho(u_1)^2\, dy - \int_0^w \rho(u_2)^2\, dy$$

And mass conservation gives:

$$\int_{CS} \rho v \cdot n\, dA = \int_0^w -\rho u_1\, dy + \int_0^w \rho u_2\, dy = 0$$

Multiply by u_1:

$$\int_0^w -\rho(u_1)^2\, dy + \int_0^w \rho u_1 u_2\, dy = 0$$

Combing these relations:

$$F_D = \int_0^w \rho_2 u_2 (u_1 - u_2) dy$$

The drag coefficient is defined as:

$$C_D = \frac{D}{\frac{1}{2}\rho U_\infty^2 S}$$

Where the reference area for the cylinder is:

$$S = (Diameter \times Length)$$

These relations were later used to evaluate the drag from the measured wake velocity profiles for the cylinder, airfoil, and 2D/3D cylinder experiments.

The second experiment conducted in Part A of this lab examined Vortex Shedding behind a circular cylinder placed in the test section. Vortex shedding occurs when flow separates alternately from each side of a bluff body (an object whose shape creates a significant amount of drag), producing an oscillating wake and a corresponding unsteady force on the object. When shedding is periodic, the shedding frequency f is related to the flow velocity v, cylinder diameter D, and Strouhal number St:

$$f = St\frac{v}{D}$$

In this experiment, a small cylinder instrumented with an internal accelerometer was mounted in the wind tunnel, and data was collected at fan settings of 30 Hz and 45 Hz. For each fan speed, three time-series samples were recorded. The acceleration signal from the vibrating cylinder captures the alternating forces generated by the vortices in its wake. By applying a Fourier transform to these signals, the dominant frequency peak corresponding to vortex shedding can be identified.

Figure 1: from researchgate.net [2]

Figure 1 shows vortex shedding under laminar flow, but the flow in our test conditions transitions between laminar and turbulent depending on the Reynolds number. As a result, the wake structure and shedding frequency vary with fan speed.

Vortex shedding is a heavily considered phenomenon in engineering. Tall buildings, towers, and long-span bridges must be evaluated for vortex-induced vibrations, because if the shedding frequency approaches the natural frequency of the structure, dangerous resonant oscillations may occur. The Tacoma Narrows Bridge collapse is a well-known historical example of the instability influenced by unsteady aerodynamic loading. [3]

Beyond civil structures, vortex shedding plays an important role in many other real world applications. For instance: Offshore wind turbines experience periodic loading from shedding, influencing fatigue life. Shedding in power-line cables can induce “galloping,” leading to larger-amplitude oscillations.

Understanding the relationship between flow velocity, Strouhal number, and shedding frequency is therefore essential for predicting aerodynamic excitation and ensuring that structures and components can withstand unsteady fluid forces.

In the third experiment of Part A of this lab, we examined the wake structure behind a circular cylinder to quantify drag forces through velocity field measurements. When flow encounters a bluff body like a cylinder, the boundary layer separates from the surface, creating a low velocity wake region downstream. This wake is characterized by a momentum deficit, the flow velocity behind the cylinder is significantly reduced compared to the free stream velocity.

To measure this effect, a pressure rake was positioned downstream of the cylinder at a distance of eight diameters, where the static pressure in the wake has recovered to approximately freestream conditions. By comparing velocity profiles with and without cylinders present, we can calculate the drag force using control volume analysis and the integral moment equation derived earlier. The drag per unit span is determined by integrating the momentum deficit across the wake.

$$D' = \int \rho u_2 (u_1 - u_2) dy$$

Where u_1 represents the freestream velocity and u_2 represents the wake velocity at each measurement location. From this drag force, we calculated the drag coefficient, which allows comparison with theoretical values. For a circular cylinder at our Reynolds numbers, the theoretical drag coefficients are found from published literature. This experiment connects the theoretical momentum conservation principles to measurable aerodynamic forces, demonstrating how wake velocity profiles directly relate to drag on bluff bodies.

In the first experiment in Part B, a boundary layer experiment was conducted using a wind tunnel and pressure tap array. The boundary layer is a thin layer of fluid next to a solid surface where viscous forces are significant. Within this layer, the velocity of the fluid gradually increases from zero at the surface (due to no-slip condition) toward the freestream velocity, where viscous effects are negligible.

In laminar flow, the velocity profile within the boundary layer is smooth and predictable, with fluid moving in orderly layers. In contrast, turbulent flow exhibits chaotic, fluctuating motions near the surface, resulting in a thicker and less predictable velocity profile.

During the experiment, velocity measurements were collected along the boundary layer at fan frequencies of 25 Hz and 45 Hz. As the freestream velocity increases, the flow tends to transition from laminar to turbulent, thickening the boundary layer. Regardless of the flow regime, the velocity at the surface remains zero, consistent with no-slip condition.

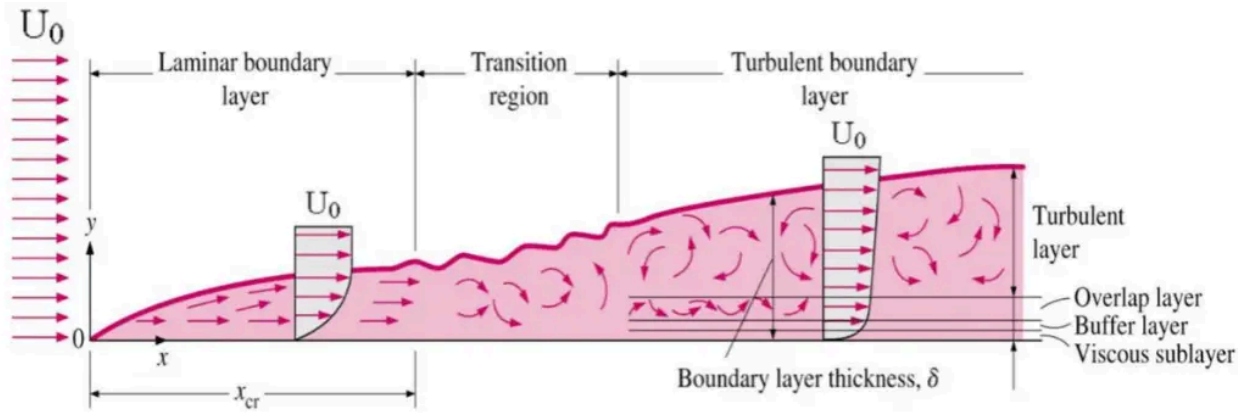

Figure 2: Lab 5 Lecture Slides [1]

The behavior illustrated in Figure 2 is consistent with my statements above, showing typical velocity distribution within the boundary layer for both laminar and turbulent flows. Observing how the boundary layer changes with position and flow speed provides insight into shear stress, drag, and overall aerodynamic performance of surfaces in real-world applications, such as aircraft wings, turbine blades, and vehicle bodies.

In the second experiment of Part B of this lab, we investigated the aerodynamic performance of a Clark-Y airfoil using a 6-DOF force balance. Unlike bluff bodies that primarily produce pressure drag, airfoils are streamlined shapes designed to generate lift efficiently while minimizing drag. The key performance metrics for an airfoil are the lift coefficient C_L, drag coefficient C_D, and pitching moment coefficient C_M, all of which vary with angle of attack.

$$C_L = \frac{L}{0.5 \cdot \rho \cdot v^2 \cdot A}$$

$$C_D = \frac{D}{0.5 \cdot \rho \cdot v^2 \cdot A}$$

$$C_M = \frac{PM}{0.5 \cdot \rho \cdot v^2 \cdot A \cdot c}$$

At low angles of attack, flow remains attached to both upper and lower surfaces of the airfoil, creating a pressure difference that generates lift. The drag in this regime is dominated by viscous forces (skin friction drag) rather than pressure drag, resulting in high lift-to-drag ratios. As angle of attack increases, lift increases approximately linearly. However, beyond a critical angle of attack, flow separation occurs on the upper surface, causing stall. At stall, lift decreases dramatically while drag increases sharply, reducing aerodynamic efficiency. [4]

To delay stall and improve high-angle-of-attack performance, we tested both “clean” and “dirty” airfoil configurations. The dirty configuration includes leading edge slats and trailing edge flaps.

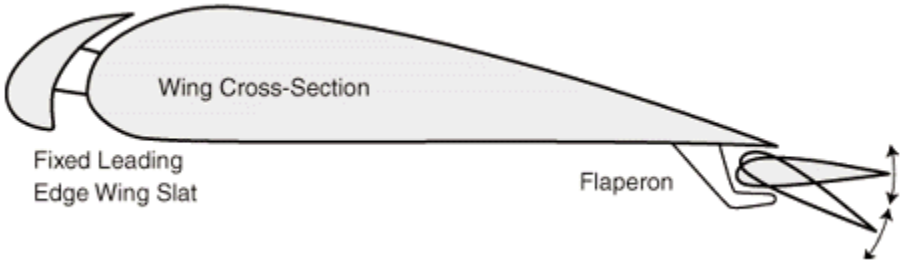

Figure 3: Airfoil with Slats and Flaps from Lab 5 Lecture Slides [1]

Slats allow high-energy air from the lower surface to re-energize the boundary layer on the upper surface, delaying separation, Flaps increase the effective camber of the airfoil, generating additional lift. These devices are commonly used during takeoff and landing when aircrafts operate at high angles of attack and low speeds.

By testing at 30 Hz and 45 Hz, we examine how flow conditions affect aerodynamic performance. Higher Reynolds numbers generally improve performance by making the boundary layer more resistant to separation, though they also increase the absolute magnitude of aerodynamic forces due to higher dynamic pressure.

In the third experiment of Part B of this lab, we investigated the difference between two-dimensional and three-dimensional flow effects on drag by testing cylinders of varying lengths. In an ideal, two-dimensional flow, the cylinder would be infinitely long, eliminating any end effects. However, real cylinders have finite length, creating three-dimensional flow patterns at the tips where trailing vortices form, similar to wingtip vortices on an aircraft.

These tip vortices represent an additional energy loss mechanism that increases the effective drag on shorter cylinders. As cylinder length increases, the proportion of the cylinder affected by three-dimensional end effects decreases, and the measured drag coefficient approaches the theoretical two-dimensional value. The longest cylinder in our experiment spans the entire width of the wind tunnel test section, touching both walls. This configuration approximates an infinite cylinder by eliminating tip losses entirely, producing the lowest drag coefficient, and most closely representing two-dimensional flow.

Understanding the distinction between two-dimensional and three-dimensional flow is critical in aerodynamic testing. Wind tunnel models are often tested with end plates or wall-to-wall configurations to simulate two-dimensional conditions, allowing researchers to isolate the aerodynamic characteristics of the cross-sectional shape without the complicating effects of finite objects. By comparing cylinders of different lengths at the same Reynolds number, this experiment quantifies how three dimensional effects increase drag as the length-to-diameter ratio decreases.

Throughout the experiments in this lab, uncertainties in measurements made were calculated using half of the resolution of the measurement tool, or using the zeroth order uncertainty calculation. These equations are provided below:

$$u_0 = \pm\frac{1}{2}resolution$$

$$u_R = \pm\sqrt{\sum_{i=1}^{L}\left(\frac{\delta R}{\delta x}\cdot u_x\right)^2}$$

3 Procedure

This lab was conducted in two sessions using a closed-loop wind tunnel. Data was collected using LabVIEW software interfaced with pressure transducers, force balances, and accelerometers. All measurements were performed at room temperature, which was recorded during tests.

Part A: Flow Characterization

Lab 5a: Equipment

- Closed loop wind tunnel
- Pitot-static tube
- Aerolab 6 Degree of Freedom Sting Balance
- Daytronic System 10 DataPac
- Pressure Systems 9010 Optomux
- 3x 2” O.D. Cylinder
- Pitot tube rake
- Thermometer
- Labview

Experiment 1: Wind Tunnel Calibration

The objective was to correlate wind tunnel fan frequency with airflow velocity for use in subsequent experiments.

1. Install pitot-static tube at tunnel centerline, approx. 16” from wall, ensuring alignment parallel to flow direction.
2. With tunnel off, record three baseline pressure readings (in Torr) and ambient temperature to establish reference
3. Start tunnel at 5 Hz and allow flow to stabilize
4. Record three pressure readings and temperature at 5 Hz
5. Increase fan frequency by 5 Hz increments, repeating measurements at each speed
6. Continue process until 40 Hz

Experiment 2: Vortex Shedding

The experiment measured the frequency of vortex shedding behind a cylinder using embedded accelerometer vibration data

1. Measure cylinder length and diameter using ruler and caliper, respectively
2. Mount accelerometer-equipped cylinder vertically in the tunnel floor at the first mounting hole
3. Set wind tunnel to 30 Hz and allow flow to stabilize
4. LabVIEW samples at 6400 Hz for 16,384 total samples
5. Collect and save one times-series data set
6. Turn the wind tunnel up to 45Hz and repeat steps.

Experiment 3: Cylinder Wake

We measured velocity profiles behind a cylinder to calculate drag force through momentum deficit analysis.

1. Measure and record vertical spacing between each pressure tap with caliper
2. Position rake vertically in tunnel
3. Set tunnel to 25 Hz and allow flow to stabilize
4. Use LabVIEW to record 20 pressure samples from each tap for averaging
5. Repeat baseline measurements at 45 Hz
6. Place cylinder 8 diameters upstream of rake and take data at 25 and 45 Hz

Part B: Aerodynamic Performance

Lab 5b: Equipment

- Closed loop wind tunnel
- Pitot-static tube
- Aerolab 6 DoF Sting Balance
- Clark Y airfoil
- Daytronic System 10 DataPac
- Pressure Systems 9010 Optomux
- Cylinders
- Pitot tube rake
- Thermometer
- Labview

Experiment 4: Boundary Layer Development

This experiment measured how the boundary layer grows along the tunnel floor by comparing velocity profiles at two streamwise locations

1. Position rake at first hole (upstream location) ensuring all pressure taps are oriented parallel to flow
2. Connect pressure tubes to scanner
3. Set wind tunnel to 30 Hz
4. Use LabVIEW to acquire pressure data
5. Manually record ambient temperature
6. Do same for 45 Hz
7. Move rake to second hole (down stream location)
8. Collect data at 30 and 45 Hz

Experiment 5: Airfoil

We measured lift, drag, and pitching moment on a Clark Y airfoil at various angles of attack in both clean and dirty(high lift) configurations.

1. Measure airfoil chord length, span, and maximum thickness with caliper
2. Mount airfoil on sting balance and secure with screws
3. Use model positioning system to set angle of attach to -10º
4. Set wind tunnel to 30 Hz
5. Launch LabVIEW and record forces and moments from six-axis balance
6. Observe wool tufts on air foil surface for signs of flow separation
7. Repeatedly increase angle of attack by 3º and collect data at each point until airfoil visibly stalls (tufts become erratic and separated from surface)
8. Take one more data 3º after stall
9. Do same steps at 45 Hz too
10. Do same steps on dirty configuration: airfoil with slats and flaps (30 and 45 Hz)

Experiment 6: 2D/3D Cylinders

This experiment compared drag on cylinders of varying length to examine three-dimensional flow effects.

1. Measure and record length and diameter of all cylinders using ruler and caliper, respectively
2. Set wind tunnel to 30 Hz
3. Run LabVIEW with no cylinder mounter to record baseline ("Cylinder 0")
4. Place each cylinder in wind tunnel and Run LabVIEW to record data at 30 Hz

4 Results

4.1 Wind Tunnel Calibration

The first portion of this lab involved calibrating the wind tunnel to establish a relationship between fan frequency and airflow velocity. The tunnel frequency was gradually increased from 0 to 40 Hz in 5 Hz increments, and the corresponding dynamic pressure was measured using a pitot-static tube. Using Bernoulli's equation, these pressure measurements were converted to velocity values. The relationship between fan frequency and air velocity is presented in Figure 4 and Table 1.

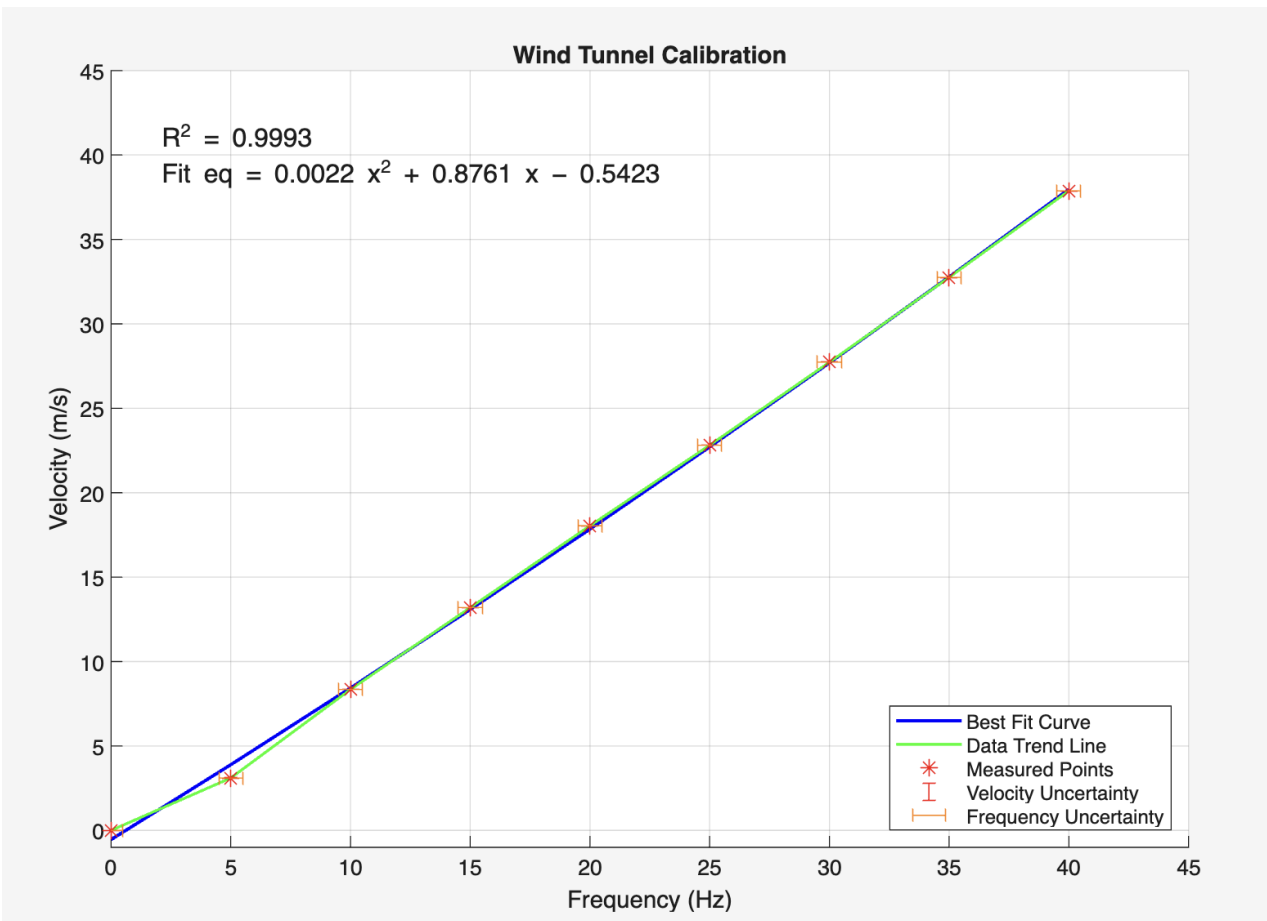

Figure 4: Wind Tunnel Calibration curve with error bars with an equation for the velocity as a function of frequency and coefficient of determination

Table 1: Velocities at different frequencies from wind tunnel calibration and uncertainties

Frequency (Hz)	Velocity (m/s)	Uncertainty - Velocity (m/s)	Uncertainty - Frequency (Hz)
0.0	0.00	± 0.000	± 0.500
5.0	3.11	± 0.0175	± 0.500
10.0	8.35	± 0.00652	± 0.500
15.0	13.20	± 0.00412	± 0.500
20.0	18.04	± 0.00302	± 0.500
25.0	22.81	± 0.00239	± 0.500
30.0	27.73	± 0.00196	± 0.500
35.0	32.75	± 0.00166	± 0.500
40.0	37.88	± 0.00144	± 0.500
40.0 (extrapolated)	37.97	n/a	± 0.500
45.0 (extrapolated)	43.27	n/a	± 0.500

This data exhibits a clear polynomial relationship between frequency and velocity. Using MATLAB's polyfit function, a second-degree polynomial was fitted to the experimental data:

$$v = 0.0022f^2 + 0.8761f - 0.5423$$

Where v is the velocity in m/s and f is the frequency in Hz. Extrapolated values on the table are calculated from this calibration equation, the very small difference in 40.0 Hz and 40.0 Hz - extrapolated prove the calibration equation is valid. Also, this fit is achieved with an R^2 value 0.9993 indicating excellent agreement between the model and the experimental data. The high R^2 value confirms that the polynomial accurately represents the relationship and can be reliably used for extrapolation.

This calibration equation was essential for determining velocities at fan frequencies beyond our direct measurement range. Specifically 45 Hz was extrapolated for use in the vortex shedding section. We needed the velocity at a fan frequency of 45 Hz for necessary calculations in that section (Strohaul number, Reynolds number). The polynomial form is physically reasonable, as it accounts for non-linear effects in fan performance and tunnel flow characteristics at higher speeds.

4.2 Vortex Shedding

This experiment examined the vortex shedding behavior of a cylinder equipped with an internal accelerometer. The cylinder was subjected to airflow at fan speeds of 30 Hz and 45 Hz, and acceleration data was recorded to capture the vibrational response induced by alternating vortex shedding. Time-domain and Frequency-domain analyses were performed to identify the dominant shedding frequency at each fan speed.

Using the empirical formula for Strohaul Number:

$$St = 0.198(1 - \frac{19.7}{Re}),$$

theoretical shedding frequencies were calculated to be 115.252 Hz at 30 Hz fan speed and 179.868 Hz at 45 Hz fan speed. These predicted frequencies represent the rate at which vortices alternately detach from opposite sides of the cylinder, creating periodic forces that cause structural vibrations.

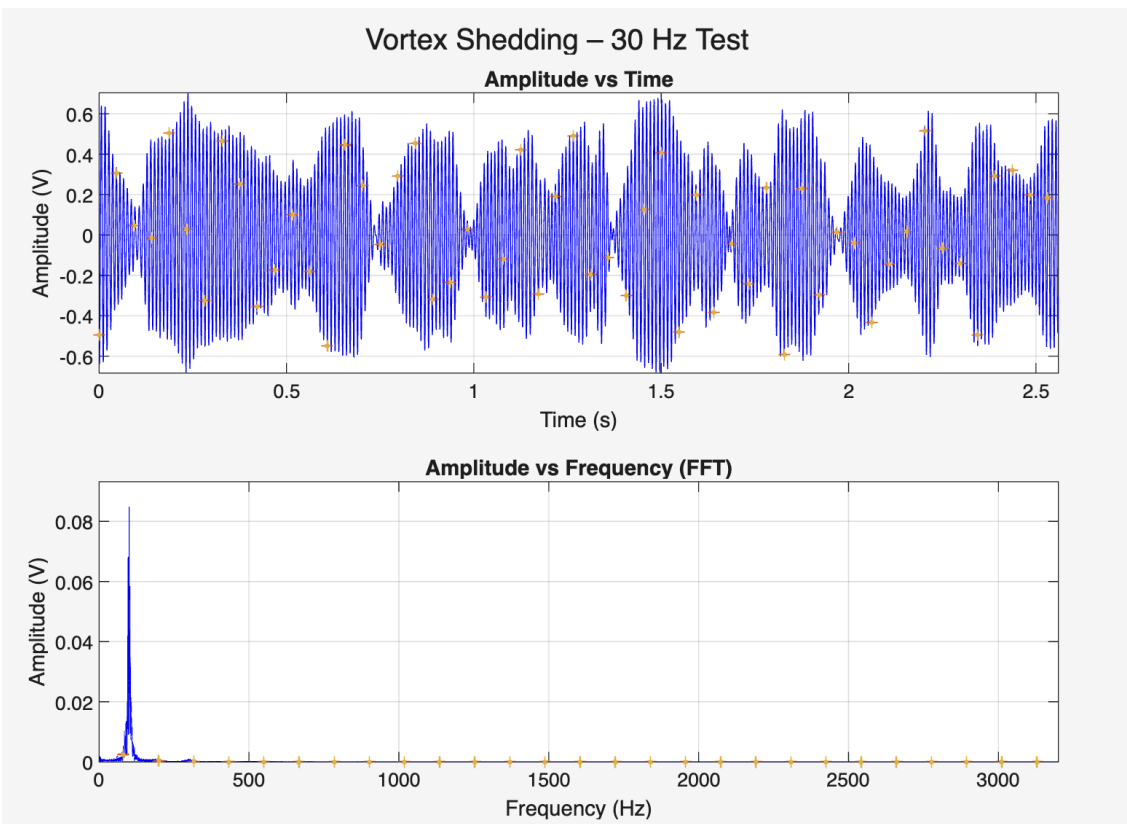

Figure 5: Amplitude vs Time and Amplitude vs Frequency subplots for 30 Hz vortex shedding experiment

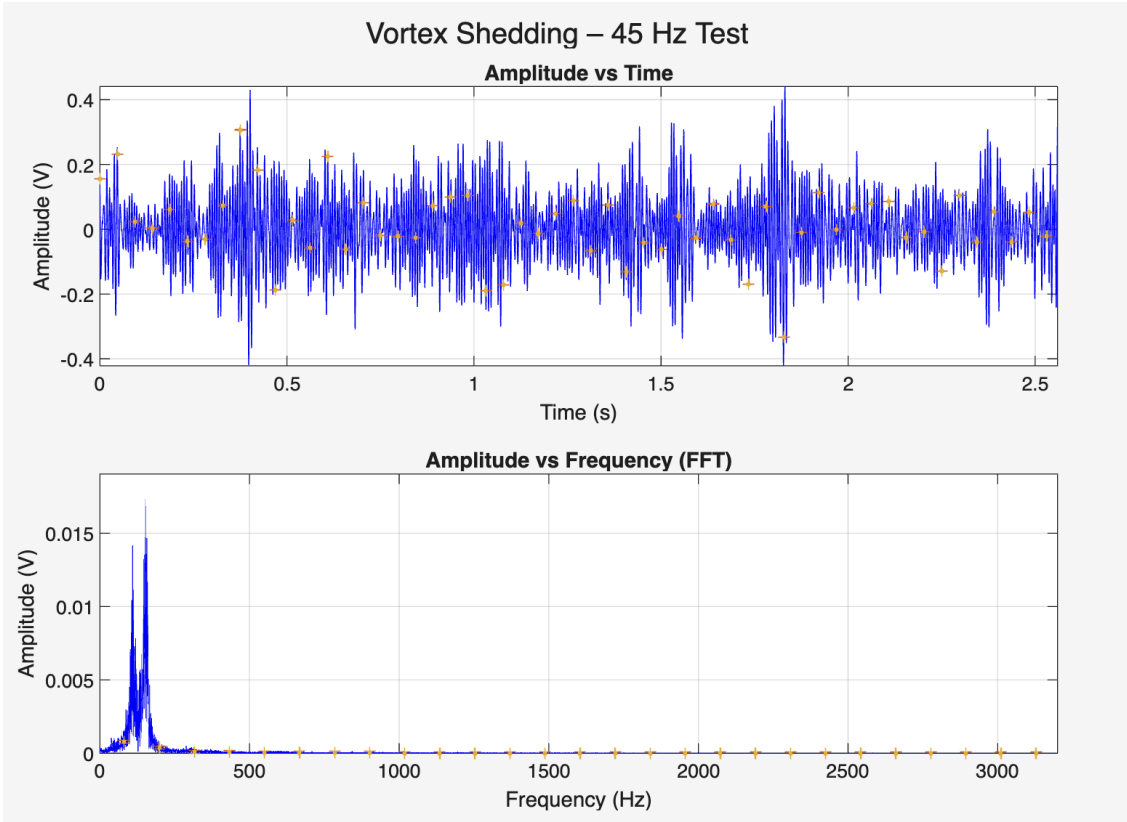

Figure 6: Amplitude vs Time and Amplitude vs Frequency subplots for 30 Hz vortex shedding experiment

The cylinder vibrated most violently at 45 Hz, as shown by the larger amplitude Figure 6. This occurs due to resonance. When the vortex shedding frequency matches the cylinders natural frequency, vibration amplitude increases dramatically. Maximum vibration does not necessarily occur at the maximum flow speed, but rather when the shedding frequency (which varies with velocity through $f_s = St \cdot v/D$) aligns with a structural natural frequency. The 45 Hz condition produced a shedding frequency near the mounted cylinders natural frequency, causing resonant amplification.

It is also notable that the same accelerometer works in both the vibrations and fluids labs because it measures structural motion regardless of excitation source. Whether the cylinder is driven by mechanical shakers or alternating aerodynamic forces from vortex shedding, the accelerometer simply records the resulting acceleration. The frequency content of this signal reveals the dominant excitation frequency: that being the natural frequencies in vibrations testing or the shedding frequency in fluid flow. The accelerometer just measures structural response allowing it to work in both labs. The quantitative results for both test conditions are summarized in Table 2.

Table 2: Frequency, velocity, Strohauls Number, Reynolds number, theoretical and experimental shedding frequencies for both 30 and 45 Hz vortex shedding experiment

	Experiment 1	Uncertainty	Experiment 2	Uncertainty
Tunnel Frequency (Hz)	30.0	± 0.500	45.0	± 0.500
Tunnel Velocity (m/s)	27.727	± 0.00196	43.269	n/a
Strohauls Number	0.198	± 3.595e-07	0.198	± 2.304e-07
Reynolds Number	9.042e+04	± 7.535e+02	1.411e+05	± 1.176e+03
Theoretical Shedding Frequency (Hz)	115.249	± 0.960	179.865	± 1.499
Experimental Shedding Frequency (Hz)	100.391	± 0.500	151.172	± 0.500

The experimental shedding frequencies were identified by locating the dominant peaks in the frequency-domain plots obtained from Fast Fourier Transform (FFT) analysis. At 30 Hz fan speed, the measured shedding frequency was 100.391 Hz, yielding a ~13% error from the theoretical prediction of 115.249 Hz. At 45 Hz fan speed, the measured frequency was 151.172 Hz, representing a ~16% error from the predicted 179.865 Hz.

Both measurements show reasonable agreement with theory, though both deviate by 13-16% from predicted values. The consistent underestimation of shedding frequency at both flow speeds suggests a systematic effect. Several factors may contribute to this discrepancy. First, the empirical Strouhal relationship $St = 0.198(1 - \frac{19.7}{Re})$ is derived from ideal conditions and may not fully capture the specific flow characteristics in our wind tunnel, including turbulence intensity, wall effects, and potential differences in boundary conditions from those used to derive the empirical formula. Second, the cylinder mounting configuration and accelerometer installation may introduce mechanical damping that affects the vibrational response potentially shifting the dominant frequency observed in the acceleration signal. Third, at these Reynolds numbers ($Re \sim 10^5$), the flow is in a transitional regime where the wake structure becomes increasingly complex, making the shedding less periodic and the dominant frequency less sharply defined.

Despite these deviations, the results clearly demonstrate the fundamental relationship between flow velocity and vortex shedding frequency: high flow speeds produce proportionally high shedding frequencies, as expected. The measured frequencies confirm that vortex-induced vibrations occur at distinct, measurable frequencies that scale predictably with flow conditions. Understanding these frequencies remains critical for structural design, and resonance between the shedding frequency and the structure's natural frequency can lead to catastrophic vibration-induced failure.

4.3 Cylinder Wake

This experiment investigated the wake structure and drag force on a cylinder by measuring velocity profiles downstream using a pressure rake. The cylinder was positioned 8 diameters upstream of the rake, and measurements were taken at fan speeds of 25 and 45 Hz. Velocity profiles were measured both with and without the cylinder present to quantify the momentum deficit in the wake, which directly relates to the drag force on the cylinder

The wake behind a bluff body is characterized by a significant velocity deficit due to flow separation and the formation of recirculating vortices. The momentum deficit creates a parabolic velocity profile with the minimum velocity occurring directly behind the cylinder centerline. As distance from the centerline increases, the wake velocity gradually recovers toward the freestream value. Higher fan speeds are expected to produce greater drag forces due to increased dynamic pressure, though the drag coefficient may vary with Reynolds number.

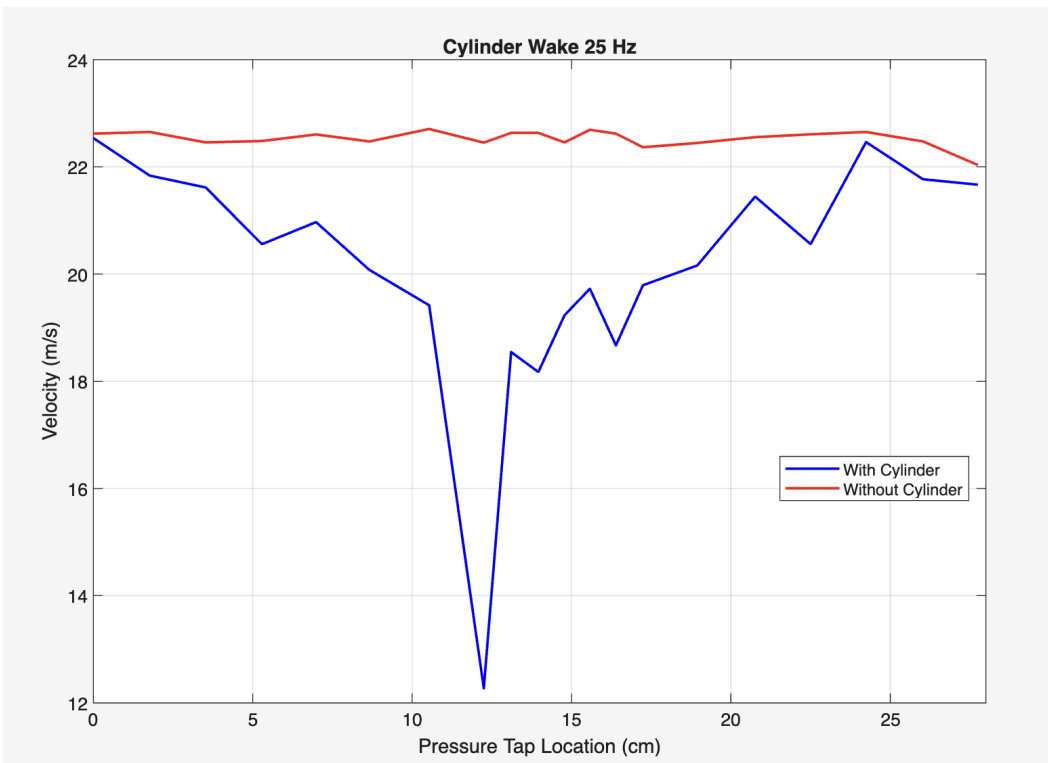

Figure 7: Velocity profile across the rake at 25 Hz fan speed, comparing freestream flow (without cylinder) to wake flow (with cylinder)

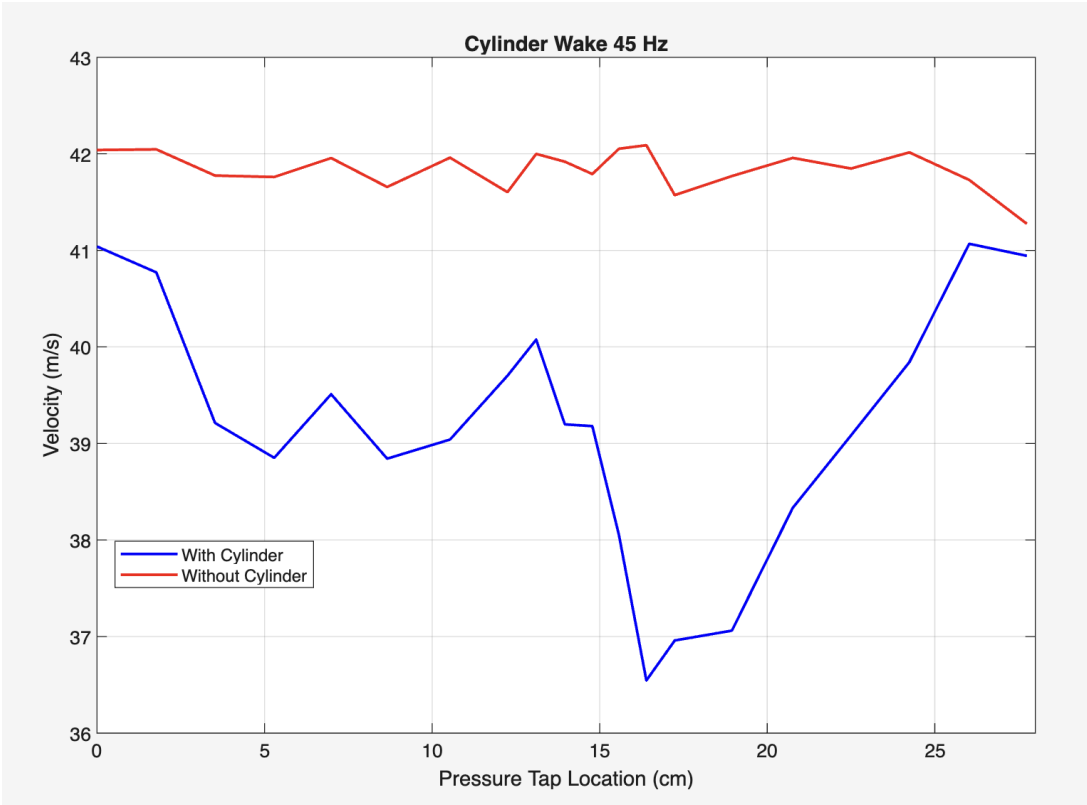

Figure 8: Velocity profile across the rake at 45 Hz fan speed, comparing freestream flow (without cylinder) to wake flow (with cylinder)

Table 3: Cylinder wake drag measurements and coefficients

	25 Hz	Uncertainty	45 Hz	Uncertainty
Tunnel Frequency (Hz)	25	± 0.500	45	± 0.500
Drag Force (N)	8.554	± 0.795	21.409	± 1.073
Drag Coefficient	0.924	± 0.128	0.670	± 0.077
Reynolds number	7.519e+04	± 7.715e+03	1.396e+05	± 1.433e+04

The velocity profiles in Figure 7 and 8 clearly show the characteristic parabolic wake structure behind the cylinder with minimum velocity occurring at the centerline where flow separation is most pronounced. The wake width and depth both increase with fan speed, reflecting the higher momentum deficit at higher flow velocities. Using the integral momentum equation, drag forces were calculated by integrating the velocity deficit across the rake. At 25 Hz, the measured drag was 8.554 N, while at 45 Hz it increased to 21.409 N, which is an expected increase as drag scales with the square of velocity ($D' \propto v^2$).

While the drag force increases significantly with velocity, the drag coefficient decreases from 0.924 at 25 Hz to 0.670 at 45 Hz. This inverse relationship between drag coefficient and velocity is consistent with theory, as Cd = Fd / ((½)(ρ)(v^2)(A)) shows the coefficient is inversely proportional to v^2 (and Fd increases at a smaller rate than v^2). The Reynolds numbers for these tests were 7.519e+04 at 25 Hz and 1.396e+05 at 45 Hz.

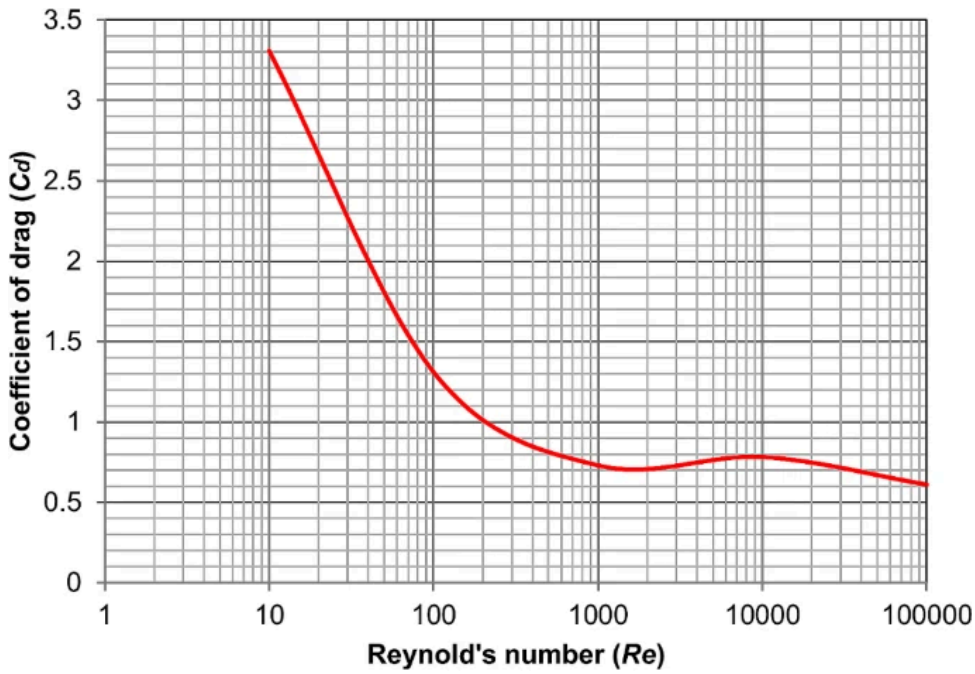

Figure 9: Drag coefficient of a circular cylinder from link.springer.com [5]

As demonstrated in Figure 9, published data for circular cylinders shows $C_d \approx 0.8$ at $Re \approx 7.5 \times 10^4$ and $C_d \approx 0.6$ at $Re \approx 1 \times 10^5$. Our measured drag coefficients of 0.924 at 25 Hz and 0.670 at 45 Hz shows reasonable agreement with these published values, with our results only approximately 10-15% higher, but within uncertainty values.

The slight overestimation at both Reynolds numbers may be attributed to several factors. First, three-dimensional flow effects near the cylinder ends can increase the effective drag beyond what is captured in idealized flow data. Second, the limited measurement resolution across the wake may introduce errors when numerically integrating the momentum deficit. Third, the finite length of our cylinder and its mounting configuration introduce boundary effects not present in published data, which typically represents infinite cylinders or cylinders with end plates.

Importantly, both our measurements and the published data demonstrate the characteristic decrease in drag coefficient with increasing Reynolds number in the subcritical regime. This decrease occurs because drag force increases at a slower rate than the dynamic pressure term ($\frac{1}{2}\rho v^2$), resulting in lower coefficient at higher flow speeds, a fundamental behavior of bluff body aerodynamics in this Reynolds number range.

4.4 Boundary Layer

This experiment measured how air velocity changes with distance from the wind tunnel floor at a fan speed of 30 Hz. A boundary layer rake containing multiple pressure taps was positioned at two locations in the test section, an upstream position (1st hole, laminar) and a downstream position (2nd hole, turbulent), to observe how the boundary layer develops as flow travels along the wall. Refer to Figure 2 in the introduction to see where we placed the rake for each set of data.

The no-slip condition says that air touching the wall has zero velocity due to viscous forces between fluid layers and the solid surface. As distance from the wall increases. Velocity gradually rises until reaching freestream conditions. The boundary layer thickness δ is defined as the height where local velocity equals 99% of freestream velocity. Within this region, flow is rotational and unsteady, meaning Bernoulli's equation cannot be applied. Only the full Navier-Stokes equation describes the physics, though these are difficult to solve analytically.

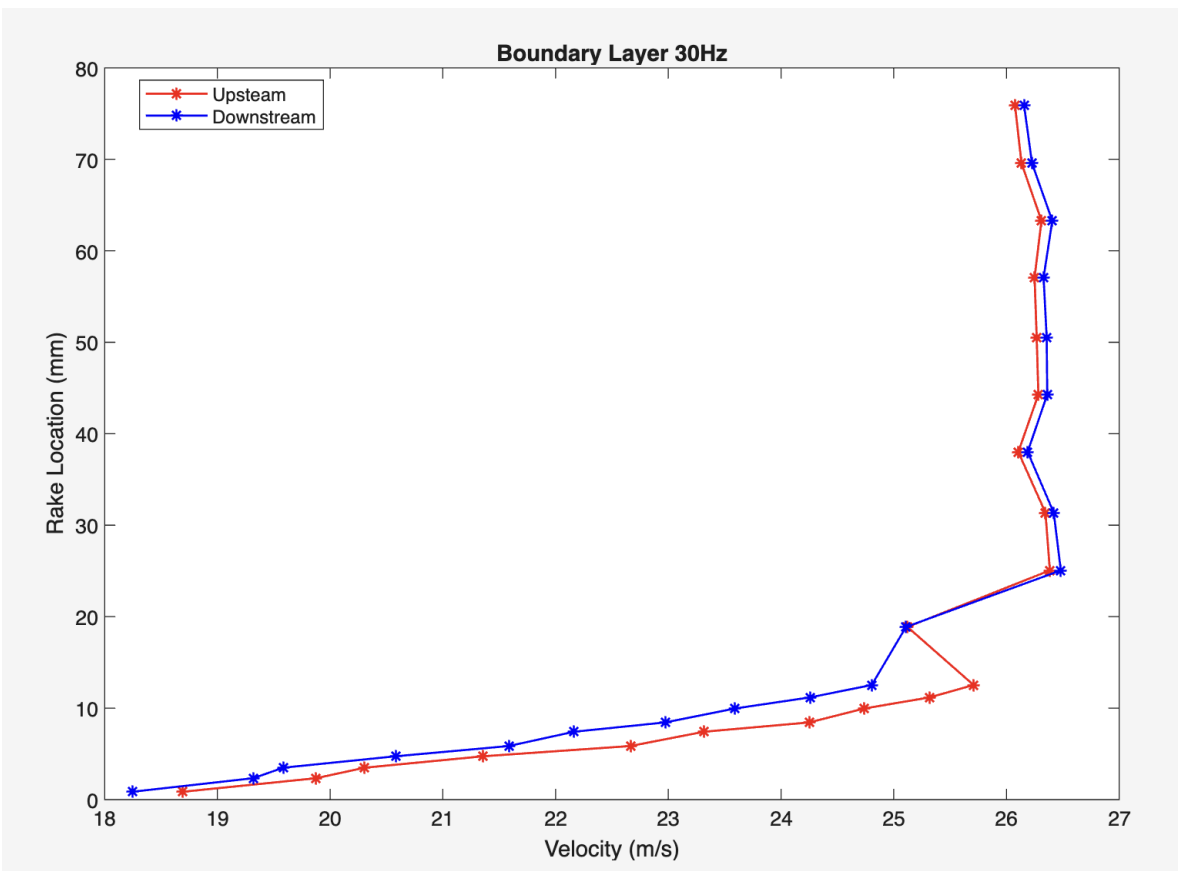

Figure 10: Velocity profile vs distance from wall at 30 Hz, comparing upstream and downstream measurement positions.

Figure 10 clearly shows that as the rake approaches the wall, the measured velocity decreases significantly. This trend reflects the no-slip condition, which requires the fluid velocity to be zero at the physical surface. In the experiment, the near-wall taps do not read exactly zero because the first tap center is located ~1.0mm above the ground and the pressure transducer has finite resolution. Nevertheless, the trend demonstrates the correct boundary-layer structure.

You'll also see from Figure 10, the velocity in the center of the test section does slightly change with distance downstream. This variation is a direct consequence of continuity and boundary-layer growth. As flow moves downstream, the boundary layer on the walls thickens, which reduces the effective flow area in the core of the test section. Because the mass flow rate must remain constant (continuity), the fluid in the remaining unblocked central region must speed up slightly to compensate for the growing low-momentum region near the walls. As a result, the centerline velocity typically increases by a small amount between the upstream and downstream measurement stations. This change is not large, because the boundary layer is still thin relative to the size of the test section, but it is measurable and entirely expected.

As you move downstream, the boundary layer steadily thickens and its profile becomes fuller, meaning the low-velocity region near the wall extends farther into the flow. This happens because viscous shear at the wall slows the fluid, and viscosity continuously spreads this momentum deficit outward, causing gradual growth of the layer. Once transition occurs, turbulent mixing greatly accelerates momentum transfer, so the boundary layer thickens even faster. The combined effect of wall friction, viscous diffusion, and (if present) turbulent mixing is what drives the downstream evolution of the boundary layer profile.

4.5 Airfoil

This experiment measured the aerodynamic forces and moments on a Clark Y airfoil using a six-degree-of-freedom force balance. Testing was conducted at two flow speeds (30 Hz and 45 Hz) in both clean configuration and dirty configuration (with leading-edge slats and trailing-edge flaps deployed). The angle of attack was varied from approximately -10° through stall in ~3° increments to characterize lift, drag, and pitching moment behavior across the operational range. Forces measured by the balance were resolved into lift F_L and drag F_D components using:

$$F_L = F_{Norm} \cdot cos(\alpha) - F_{axial} \cdot sin(\alpha)$$

$$F_D = F_{Norm} \cdot sin(\alpha) + F_{axial} \cdot cos(\alpha)$$

Equations found via aerospaceweb.org [6]

Where F_{Norm} is the normal force, F_{axial} is the axial force and α is the angle of attack. Aerodynamic coefficients were calculated as:

$$C_L = \frac{L}{0.5 \cdot \rho \cdot v^2 \cdot A}$$

$$C_D = \frac{D}{0.5 \cdot \rho \cdot v^2 \cdot A}$$

$$C_m = \frac{PM}{0.5 \cdot \rho \cdot v^2 \cdot A \cdot c}$$

Where A is the platform area ($Chord \times Span$) and c is the chord length.

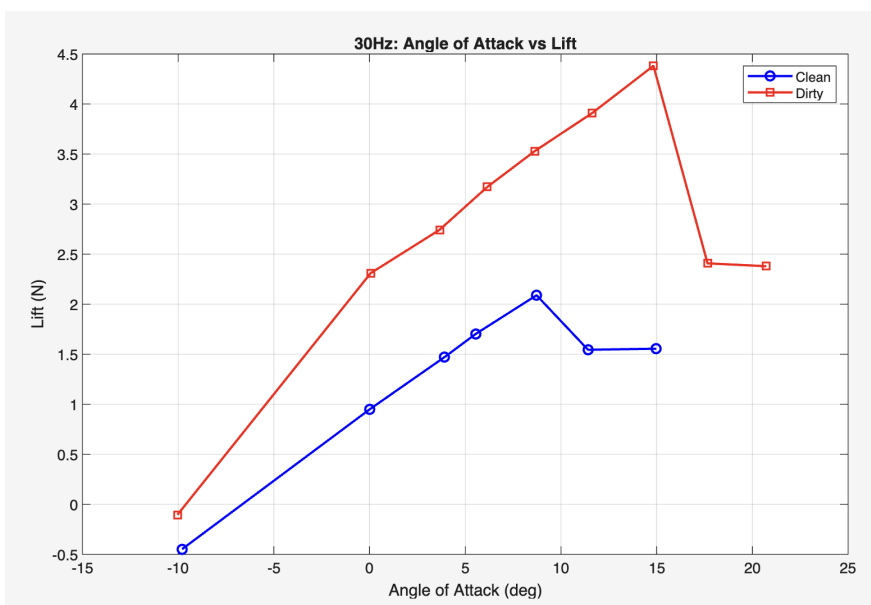

Figure 11: Angle of attack vs Lift at 30 Hz clean and dirty configurations

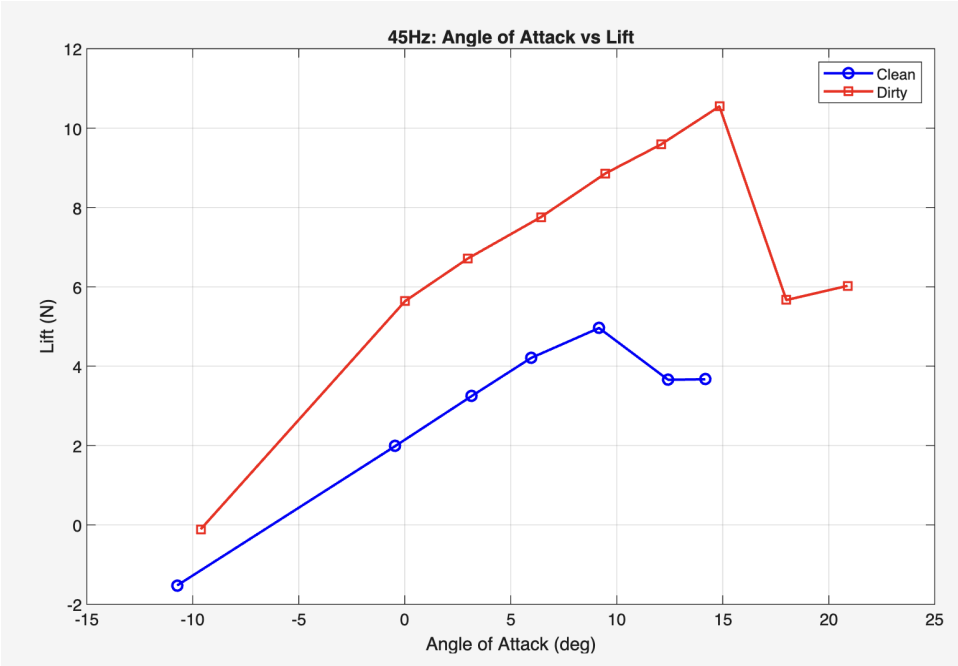

Figure 12: Angle of attack vs Lift at 45 Hz clean and dirty configurations

Figures 11 and 12 show lift increases linearly until stall at ~9° (clean) and ~15° (dirty). Slats delay stalls by ~6° as they reenergize the boundary layer. Absolute lift forces at 45 Hz are 2.4 times higher than at 30 Hz, matching the $(43.3/27.7)^2 = 2.44$ dynamic pressure ratio. Reynolds number is independent in this stall behavior range as stall angle remains consistent at both Reynolds numbers.

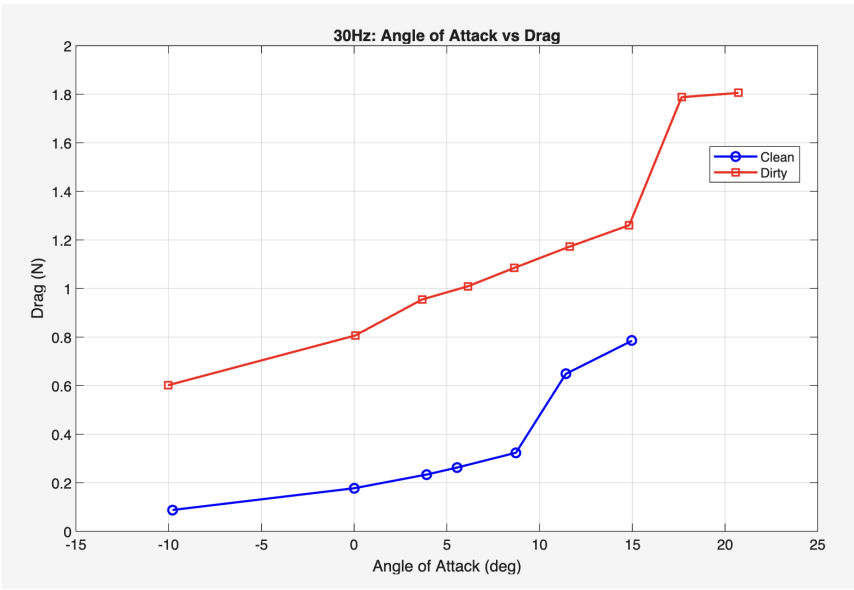

Figure 13: Angle of attack vs Drag at 30 Hz clean and dirty configurations

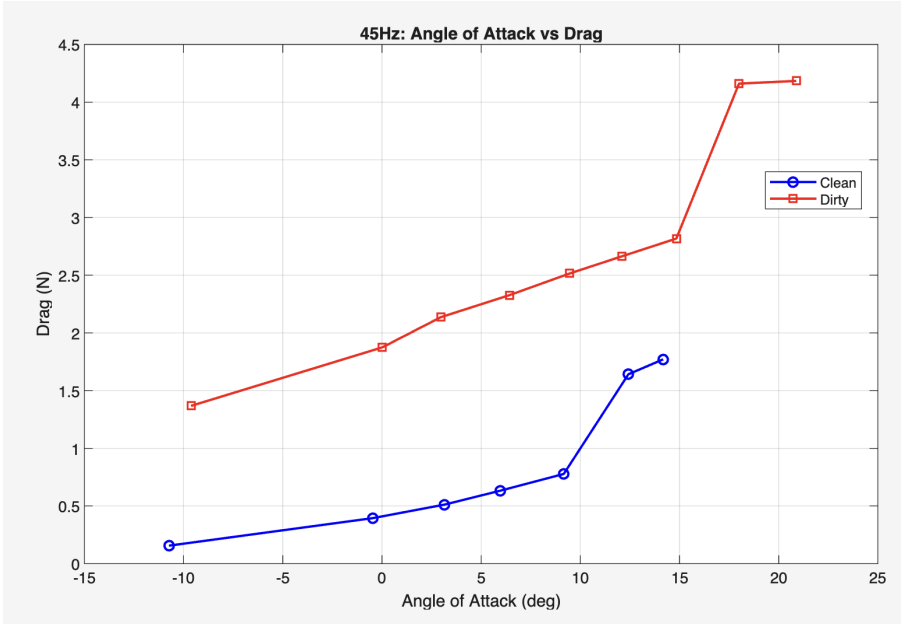

Figure 14: Angle of attack vs Drag at 45 Hz clean and dirty configurations

Figures 13 and 14 show drag remains low at small stall angles, then rises steeply near stall as pressure drag from flow separation increases. Dirty configurations experience moderately higher drag due to additional surface area from flaps and slats, though this is minor compared to lift gains. Drag at 45 Hz is 2.4x higher than that at 30 Hz, consistent with dynamic pressure scaling. Clean configurations show (partially) sharper drag rise at stall due to sudden separation, while dirty configuration separation is more gradual because surface roughness causes earlier, more gradual flow separation.

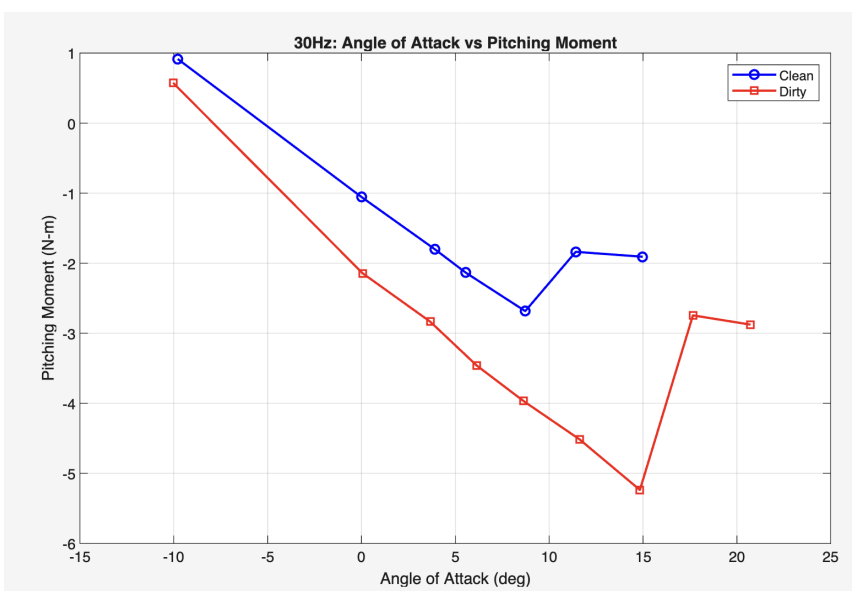

Figure 15: Angle of attack vs Pitching Moment at 30 Hz clean and dirty configurations

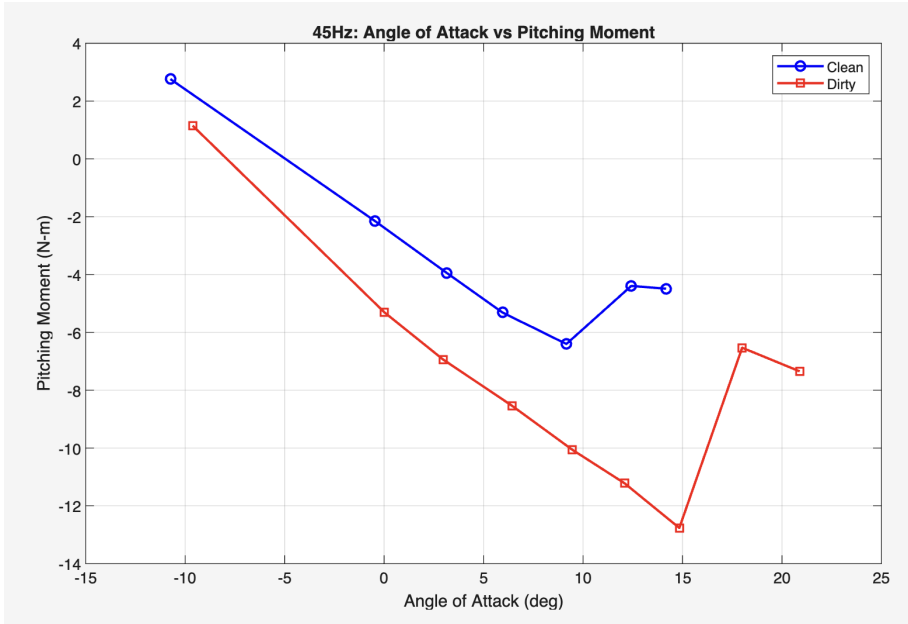

Figure 16: Angle of attack vs Pitching Moment at 45 Hz clean and dirty configurations

Figures 15 and 16 show pitching moment becomes increasingly negative as angle of attack increases, characteristic of cambered airfoils where the center of pressure moves back with increasing lift, because as the lift acts further behind the main point of rotation, it creates a stronger nose-down turning force. Dirty configuration produces larger magnitude moments due to increased lift from flaps creating additional camber. Moments at 45 Hz are approximately 2.4x larger than those at 30 Hz, reflecting correct dynamic pressure scaling.

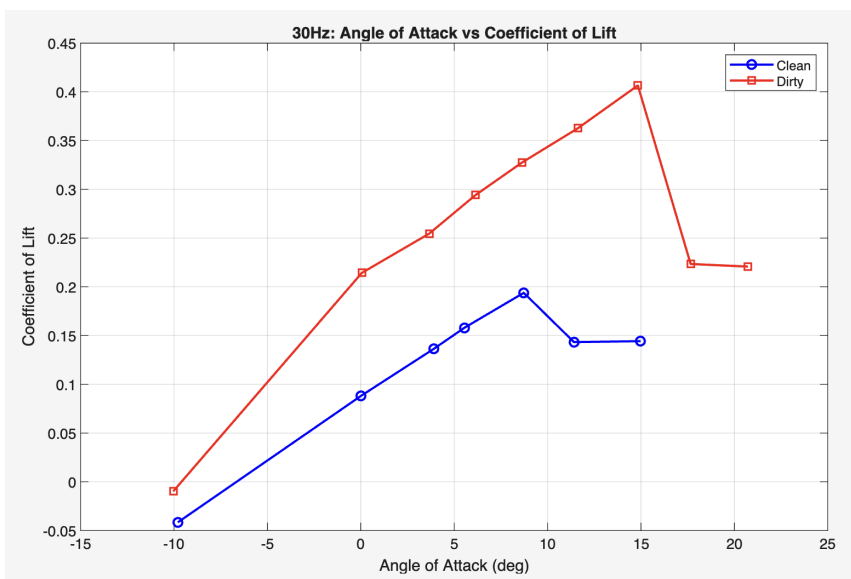

Figure 17: Angle of attack vs Coefficient of Lift at 30 Hz clean and dirty configurations

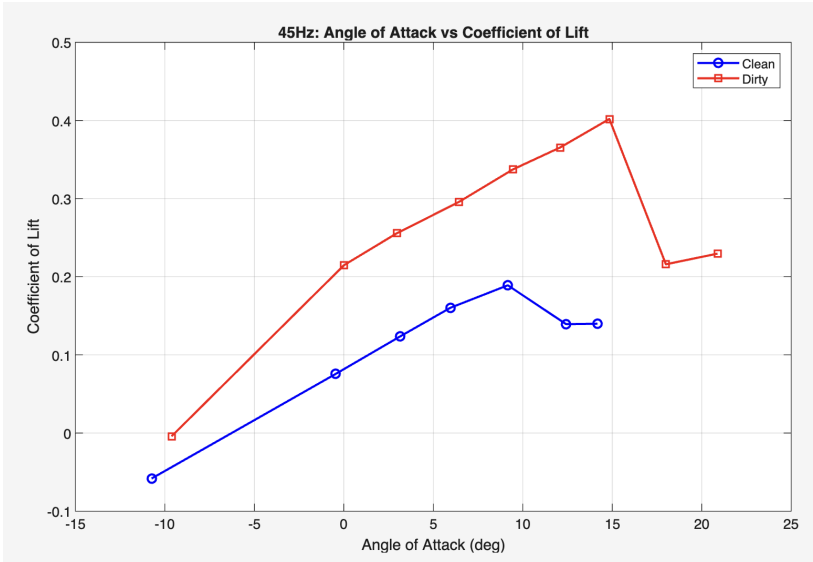

Figure 18: Angle of attack vs Coefficient of Lift at 45 Hz clean and dirty configurations

The lift coefficient normalizes data by dynamic pressure and area. From Figures 17 and 18, the slope of the linear region of the clean configuration airfoils are the lift-curve slopes, they are 0.728 rad^{-1} at 30 Hz and 0.712 rad^{-1} at 45 Hz. Thin airfoil theory predicts $2\pi \approx 6.28$ rad^{-1} for infinite span; my values are approximately 11-12% of theory due to three-dimensional wingtip effects and induced drag. The close agreement between slopes (0.728 vs 0.712) confirms this parameter is not affected by flow speed (Reynolds number) in the subcritical range, because before stall the airflow behaves similarly at both speeds, so the initial lift response stays nearly the same. Dirty configurations achieve higher peak C_L due to enhanced circulation from high-lift devices.

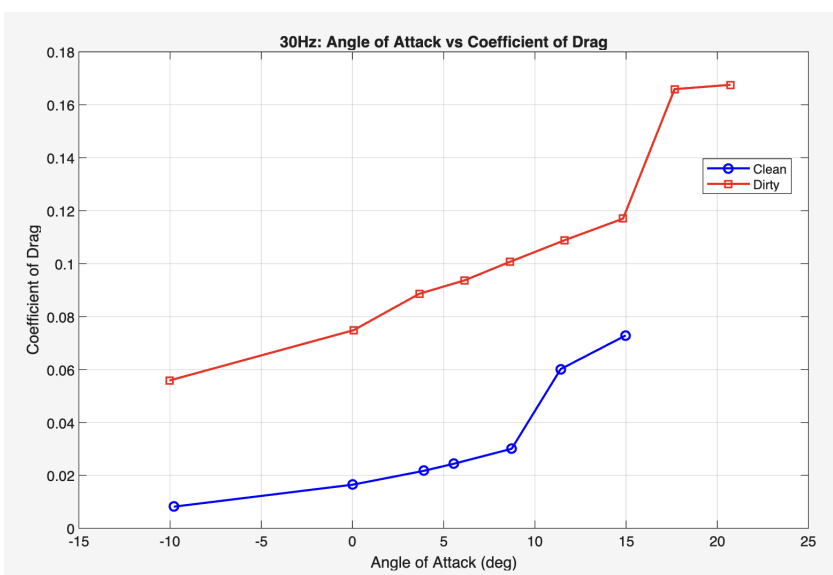

Figure 19: Angle of attack vs Coefficient of Drag at 30 Hz clean and dirty configurations

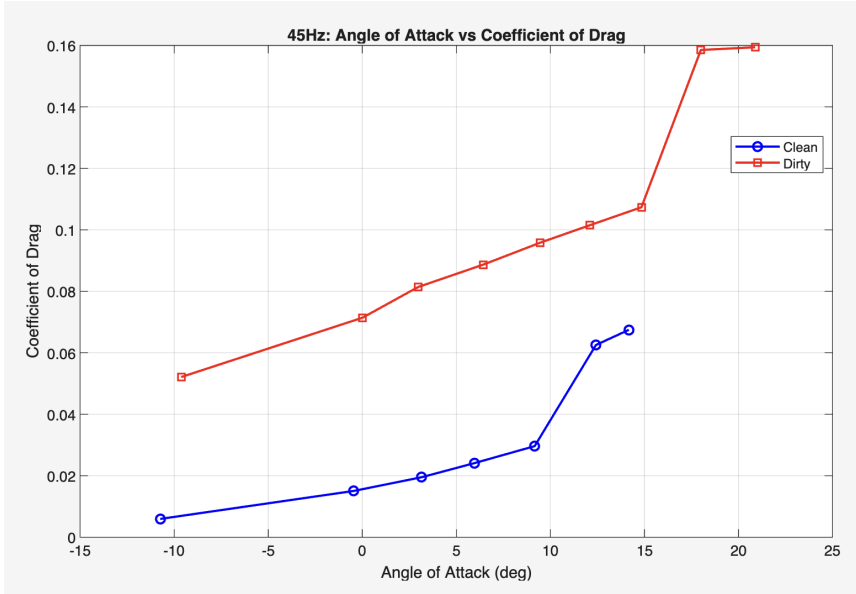

Figure 20: Angle of attack vs Coefficient of Drag at 45 Hz clean and dirty configurations

The drag coefficient C_D represents the aerodynamic performance normalized by dynamic pressure and reference area. Figures 19 and 20 show that both clean and dirty configurations show a gradual increase in C_D at low angles of attack, followed by a sharper rise near stall. The clean configurations show lower drag coefficient than the dirty configurations across the full range of angles of attack, due to added drag from rough surfaces or high-lift devices (slats/flaps) in the dirty setup. Overall, 30 Hz and 45 Hz curves are similar, demonstrating successful Reynolds normalization (drag behavior consistent across flow speeds).

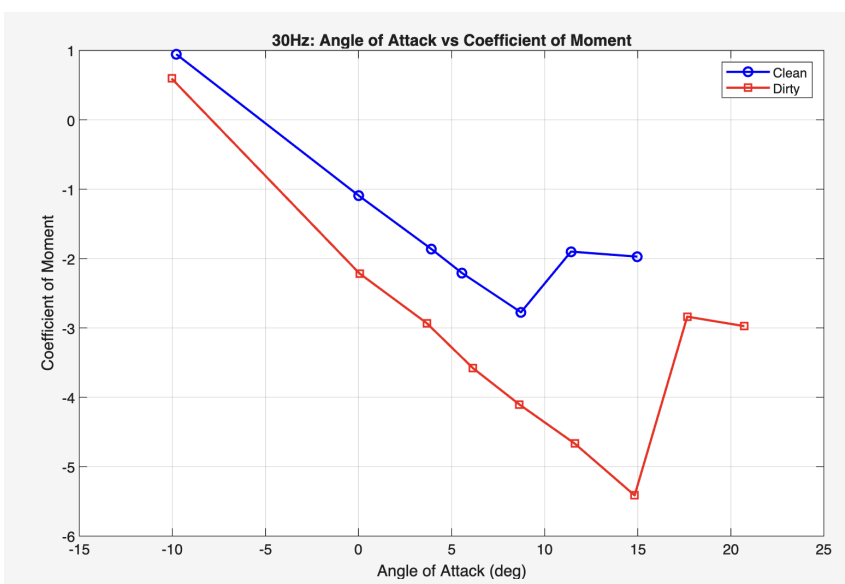

Figure 21: Angle of attack vs Coefficient of Pitching Moment at 30 Hz clean and dirty configurations

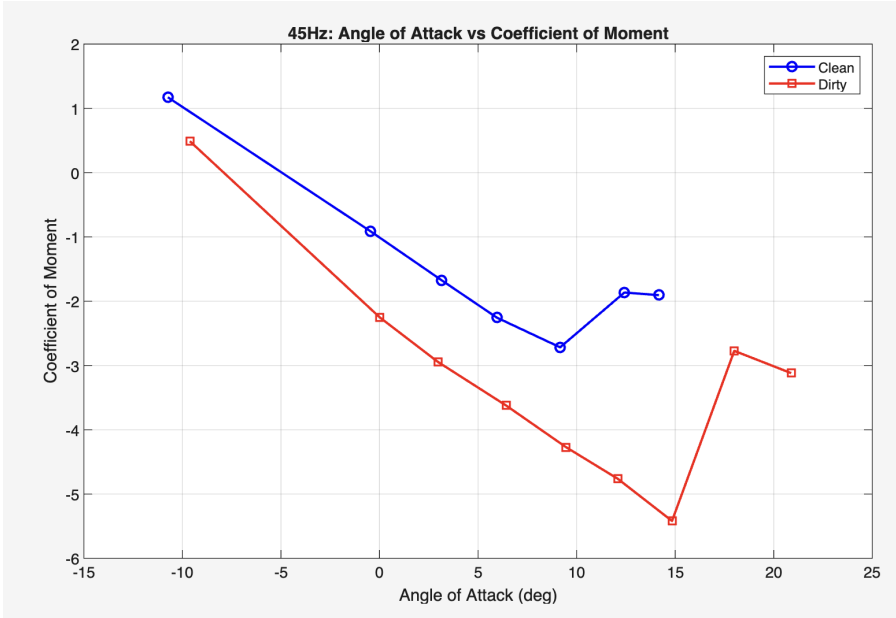

Figure 22: Angle of attack vs Coefficient of Pitching Moment at 45 Hz clean and dirty configurations

Figures 21 and 22 show that coefficient of pitching moment C_M decreases approximately linearly with angle of attack in the pre-stall region. Dirty configuration produces more negative C_M due to additional lift generated behind the aerodynamic center, likely from trailing-edge flow disturbances. The curves for 30 Hz and 45 Hz are nearly identical, indicating successful Reynolds number normalization and demonstrating that the behavior is dominated by airfoil geometry rather than Reynolds effects.

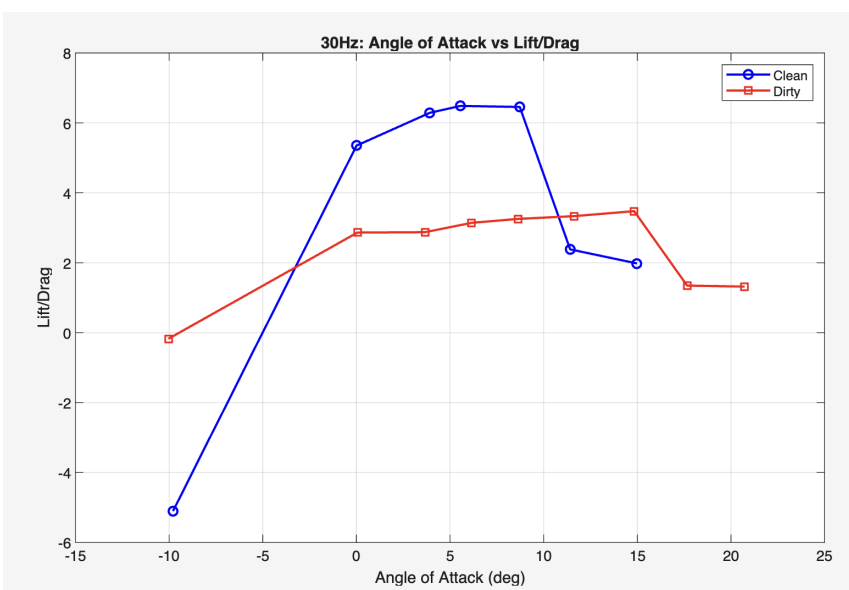

Figure 23: Angle of attack vs Lift-to-Drag ratio at 30 Hz clean and dirty configurations

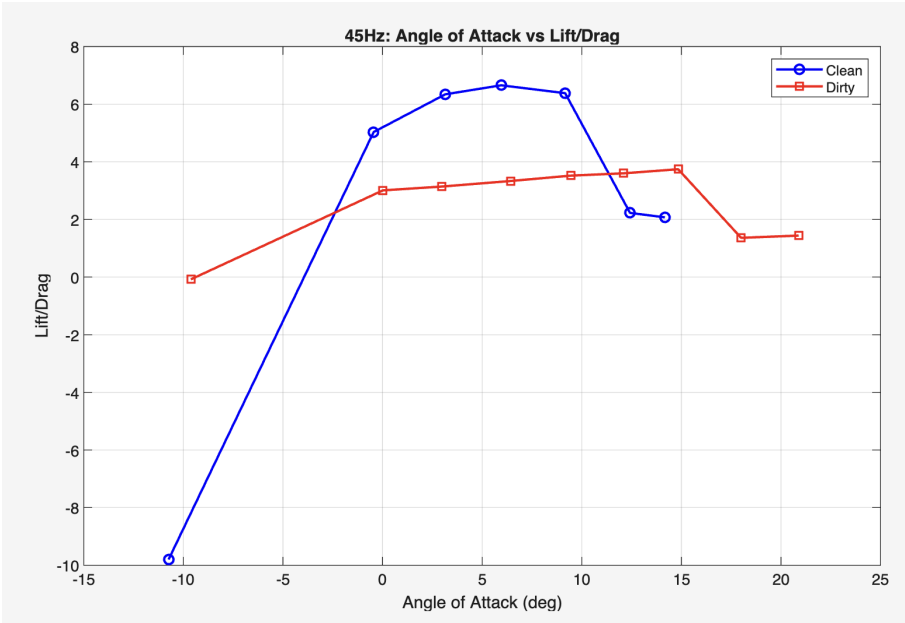

Figure 24: Angle of attack vs Lift-to-Drag ratio at 45 Hz clean and dirty configurations

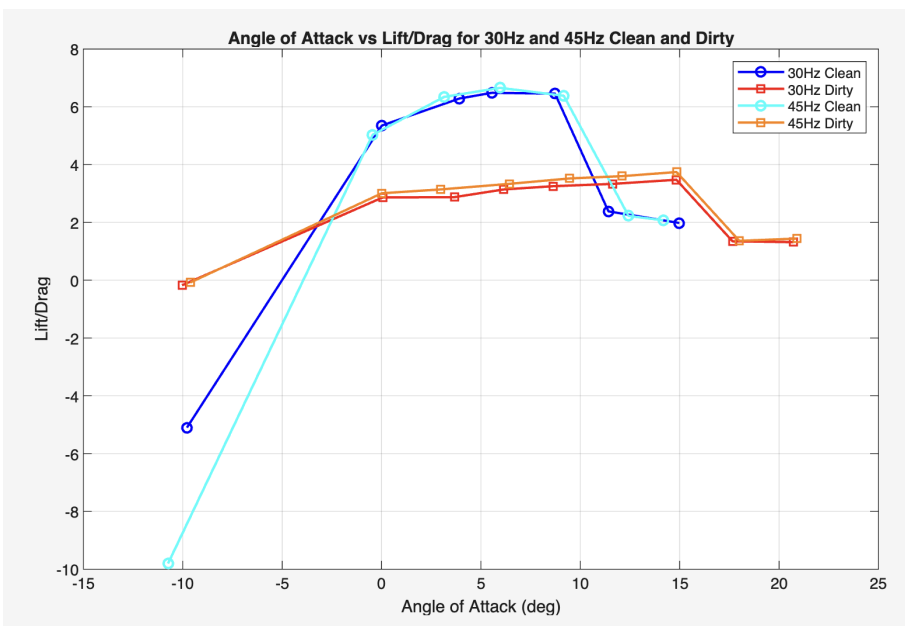

Figure 25: Angle of attack vs Lift-to-Drag ratio for all test conditions (30 Hz and 45 Hz, clean and dirty)

From Figures 23-25, I found that clean configuration achieves L/D peak of 6.49 at 30 Hz and 6.66 at 45 Hz, while dirty configurations reach maximum L/D of 3.48 at 30 Hz and 3.74 at 45 Hz. Figure 25 reveals the critical advantage: clean efficiency collapses around 9º due to stall, while dirty configurations maintain positive L/D to ~15º. Above ~10º, dirty actually outperforms clean because clean has stalled. This extended operational area explains why aircrafts deploy flaps during takeoff and landing when high angles at low speeds are necessary, the loss of efficiency is acceptable compared to stalling benefits.

Table 4: Airfoil Aerodynamic Performance Parameters at 30 Hz and 45 Hz, Clean and Dirty

	30 Hz – Clean	30 Hz – Dirty	45 Hz – Clean	45 Hz – Dirty
Tunnel Frequency (Hz)	30.0	30.0	45.0	45.0
Tunnel Velocity (m/s)	27.727	27.727	43.269	43.269
Reynolds Number	1.684e+5	1.684e+5	2.628e+5	2.628e+5
Peak Efficiency (L/D) – Clean	6.49	–	6.66	–
Peak Efficiency (L/D) – Dirty	–	3.48	–	3.74
Lift-Curve Slope (per deg)	0.0127	–	0.0124	–
Lift-Curve Slope (per rad)	0.728	–	0.712	–
Stall Angle (deg)	~9°	~15°	~9°	~15°

The measured lift-curve slopes for the clean airfoil configurations are 0.728 rad^{-1} at 30 Hz and 0.712 rad^{-1} at 45 Hz. Compared to thin airfoil theory, which predicts a slope of 2π rad^{-1} (~6.28 rad^{-1}) for an infinite span airfoil, these values are roughly 11-12% of the theoretical value. This difference is expected due to finite wing three-dimensional effects and induced drag, and the similar slopes at both frequencies indicate that the initial lift response is not sensitive to Reynolds number in the subcritical flow range.

During the experiment, tufts on the airfoil visually separated at approximately 9° for the clean configuration and 15° for the dirty configuration, which agrees well with the stall angles indicated by the lift and lift-to-drag curves. Tuft separation occurs because the boundary layer loses momentum and detaches from the surface, creating a sudden increase in flow separation and pressure drag, this is stalling. The presence of slats and flaps in the dirty configuration energizes the boundary layer, delaying separation and increasing the stall angle.

Reynolds number effects were minimal within the tested range with Re = 1.68e+5 at 30 Hz and Re = 2.628e+5 at 45 Hz. Normalized aerodynamic coefficients, C_L, C_D, and C_M, and stall angles remained nearly identical, confirming that aerodynamic behavior is primarily dictated by airfoil geometry rather than Reynolds number. Absolute lift, drag, and pitching momentum increased with dynamic pressure, which explains the approximately 2.4x higher forces at 45 Hz compared to 30 Hz.

The freestream velocity of the wind tunnel did not significantly change with angle of attack. While local accelerations occur around the airfoil due to flow deflection and induced velocities, the overall bulk flow rate is conserved by continuity, and any local variations are accounted for through normalization in the aerodynamic coefficients

The experimental results demonstrate that the Clark Y airfoil behaves consistently with expectations for a finite, cambered airfoil.. Clean configurations provided higher efficiency (L/D ≈ 6.5), but limited angle of attack range, stalling at approximately 9º. Dirty configurations sacrificed efficiency (L/D ≈ 3.5) but extended the usable angle of attack range to ~15º, 67% improvement. Lift curve slopes were consistent across speeds, stall behavior was well predicted by tuft observations, and Reynolds number variations had negligible effects on normalized aerodynamic performance. Overall, high-lift devices effectively trade cruise efficiency for enhanced low-speed operational capability.

4.6 2D/3D Cylinders

Drag measurements were obtained for four configurations at tunnel frequency 30 Hz: a baseline case (Cylinder 0, no cylinder), and three cylinders of decreasing length. The corresponding Reynolds numbers among cylinders 1, 2, and 3 were nearly identical, reflecting similar freestream velocity and only minor differences in diameter. No Reynolds number was calculated for the baseline test because no characteristic length was present. Drag force and coefficient of drag were calculated using force balance measurements, with an assumed force uncertainty of ±0.2%

Table 5: Length, Reynolds number, Drag, and Coefficient of Drag with associated uncertainties for each cylinder

	Cylinder 0 – No cyl	Cylinder 1 – Long	Cylinder 2 – Medium	Cylinder 3 – Short
Length (m)	0.000	0.607	0.507	0.267
Reynolds Number	n/a	9.039e+4	9.077e+4	9.004e+4
Uncertainty in Reynolds Number	n/a	± 11.35	± 11.37	± 11.34
Drag (N)	0.002072	3.746	2.995	1.473
Uncertainty in Drag (N)	± 4.143e-05	± 0.07492	± 0.05991	± 0.02947
Coefficient of Drag	n/a	0.2721	0.2594	0.2442
Uncertainty in Coefficient of Drag	n/a	± 0.00544	± 0.00519	± 0.00488

According to theory, an object in two-dimensional flow experiences less drag than a three-dimensional object because a three-dimensional object experiences boundary layer separation not only on the top and bottom surfaces but also along the sides, producing additional pressure drag. Energy is also lost at the 3D tips in the form of trailing vortices. In contrast, a two-dimensional object only separates at the top and bottom surfaces, resulting in less total pressure drag, despite its larger frontal area. However, as you can see from Table 5, the measured drag forces and coefficients in this experiment show that the longest cylinder, which spans the full width of the wind tunnel and approximates two-dimensional flow, has the highest drag (3.746 N) and coefficient of drag (0.2721). The medium (3D) cylinder has a drag of 2.995 N and C_D of 0.2594, and the shortest (3D) cylinder has a drag of 1.473 N and a C_D of 0.2442. This

apparent discrepancy of theory can be described from the frontal areas of the cylinders. The longest cylinder has the largest surface area exposed to the flow, giving it more resistance to the airflow and increasing the measured drag. Therefore, although the 3D cylinders experience separation on the sides as predicted by theory, the increased frontal area of the longest cylinder dominates, resulting in higher overall drag.

The cylinder with the highest coefficient of drag is Cylinder 1, the longest cylinder. This occurs because drag depends on not only flow dimensionality but also on the surface area exposed to the airflow. Cylinder 1's greater length and total front area increases resistance to flow, outweighing the three-dimensional effects present in the shorter cylinders. Therefore, while 3D separation increases drag for Cylinders 2 and 3 relative to an equal length 2D object, their small area reduces the total drag compared to the longest cylinder.

Comparing 2D and 3D flow, my data supports the theoretical distinction: 2D flow occurs when an object spans the entire width of the wind tunnel, preventing flow around the sides, whereas 3D flow allows circulation around the cylindrical tips, increasing side separation and pressure drag. Cylinder 1 is considered "2D" because it spanned the entire width of the wind tunnel, with both of its ends touching the walls, preventing flow around its ends. In contrast, the shorter cylinders, 2 and 3, experience 3D flow, where separation occurs not only on the front and rear faces but also around the tips, increasing drag. In practice, the measured drag shows that 3D effects are present in the shorter cylinders, but the influence of the frontal area can counteract the expected increase.

As you can see from Table 5, the coefficient of drag of Cylinder 1 (2D) is consistent with the results from the Cylinder Wake experiment from Part A of Lab 5. In this experiment, Cylinder 1 had a C_D of 0.2721 at a Reynolds number of 9.039e+4. From the wake measurements at a similar Reynolds number (25 Hz fan speed, Re ≈ 7.519e+04), the C_D was 0.924, while at 45 Hz (Re ≈ 1.396e+05), the C_D was 0.670. Although the wake based coefficients are higher, both methods capture the momentum deficit caused by pressure drag behind a bluff body. The lower C_D value in the current 2D cylinder test reflects the effect of the longer cylinder spanning the wind tunnel, which reduces tip-induced 3D separation. Overall, the two methods produce drag coefficients that are in reasonable agreement given differences in measurement technique, Reynolds number, and geometry.

Finally, the measured drag coefficients compare reasonably well with published values for circular cylinders at similar Reynolds numbers. Typical C_D values at Reynolds numbers similar to those of Cylinders 1-3 range approximately from 0.9 to 0.1 (values determined via researchgate.net [7]). My measured coefficients are lower than this range, although this just reflects the effects of finite cylinder length and 2- vs 3-dimensional flow interactions that published values don't account for.

5 Conclusion

This laboratory investigation successfully characterized key aerodynamic principles through systematic wind tunnel testing, yielding results that demonstrate strong agreement with established fluid dynamics theory. The wind tunnel calibration established a quadratic relationship between fan frequency and flow velocity with $R^2 = 0.9993$, enabling accurate velocity determination throughout the experiments

Vortex shedding measurements revealed periodic wake oscillations at 100.391 Hz and 151.172 Hz for 30 Hz and 45 Hz fan speeds. While deviating 13-16% from theoretical predictions, the consistent underestimation suggests systematic effects from mounting configuration or transitional Reynolds number behavior. Maximum vibration at 45 Hz confirmed resonance between shedding frequency and structural natural frequency, illustrating the engineering importance of vortex-induced vibrations in structures such as bridges and offshore wind turbines.

Wake profiles yielded drag coefficients of 0.924 and 0.670 and Reynolds number of 7.519e+4 and 1.396e+5, aligning within 10-15% of published data. The inverse relationship between drag coefficient and Reynolds number confirms subcritical flow regime behavior. Boundary layer measurements revealed velocity gradients consistent with no-slip conditions and downstream thickening due to momentum diffusion and continuity effects.

Airfoil testing provided comprehensive Clark Y characterization. Clean configurations achieved lift-curve slopes of 0.728 rad^{-1} and 0.712 rad^{-1}, approximately 11-12% of theoretical predictions due to finite wing effects. Maximum lift-to-drag ratios 6.5 occurring near 6° angle of attack. Dirty configurations sacrificed efficiency (L/D ≈ 3.5) but extended stall from ~9° to 15°, demonstrating the fundamental tradeoff between cruise efficiency and low-speed capability that explains deployment of high-lift devices during takeoff and landing.

Cylinder testing showed the longest cylinder achieved the highest drag drag coefficient (0.2721) as increased frontal area dominated over three-dimensional separation effects. Throughout all experiments, uncertainties remained acceptable and Reynolds number effects proved minimal for normalized coefficients, confirming geometric factors dominated observed behavior.

Recommendations for improvement include: longer data acquisition periods for improved frequency resolution; higher resolution pressure transducers to reduce integration uncertainty; testing across broader Reynolds number ranges; implementing end plates to isolate two-dimensional effects successfully. Extending airfoil testing into deep stall would also provide complete post-stall characterization.

These experiments successfully demonstrated control volume analysis, momentum conservation, and dimensional analysis principles to quantify aerodynamic phenomena critical to aerospace, civil, and mechanical engineering design.

6 Appendix

6.1 References

1: Erdman, M (2025, Fall),
ture Slides

2: Basnet, K. (2010, December). Vortex Shedding from a circular cylinder; laminar flow vortex-street behind a circular cylinder. researchgate.net, https://www.researchgate.net/figure/ortex-shedding-from-a-circular-cylinder-laminar-flow-vortex-street-behind-a-circular_fig3_332548783

3: Wikimedia Foundation. Tacoma Narrows Bridge (1940). Wikipedia. https://en.wikipedia.org/wiki/Tacoma_Narrows_Bridge_(1940)

4: Cadence Fidelity. (2024, October 4). How an airfoil's angle of attack creates lift and drag. Cadence. https://resources.system-analysis.cadence.com/blog/msa2022-how-an-airfoils-angle-of-attack-creates-lift-and-drag

5: Singh, R., Singh, J., Kumar, R., & Gupta, M. (2024, July 6). Computational fluid dynamics of flow over a cylindrical tube for steady and transient flow conditions. Springer Nature Link. https://link.springer.com/article/10.1007/s12008-024-01970-4

6: Scott, J. (2004, September 19). Lift & Drag vs. Normal & Axial Force. Aerospaceweb.org | Ask Us - Lift & Drag vs. Normal & Axial Force. https://aerospaceweb.org/question/aerodynamics/q0194.shtml

7: Kološ, I., Michalcová, V., & Lausová, L. (2021, February 14). Numerical Analysis of Flow Around a Cylinder in Critical and Subcritical Regime. researchgate.net. https://www.researchgate.net/publication/349307623_Numerical_Analysis_of_Flow_Around_a_Cylinder_in_Critical_and_Subcritical_Regime

6.2 Plots and Tables

Wind Tunnel Calibration

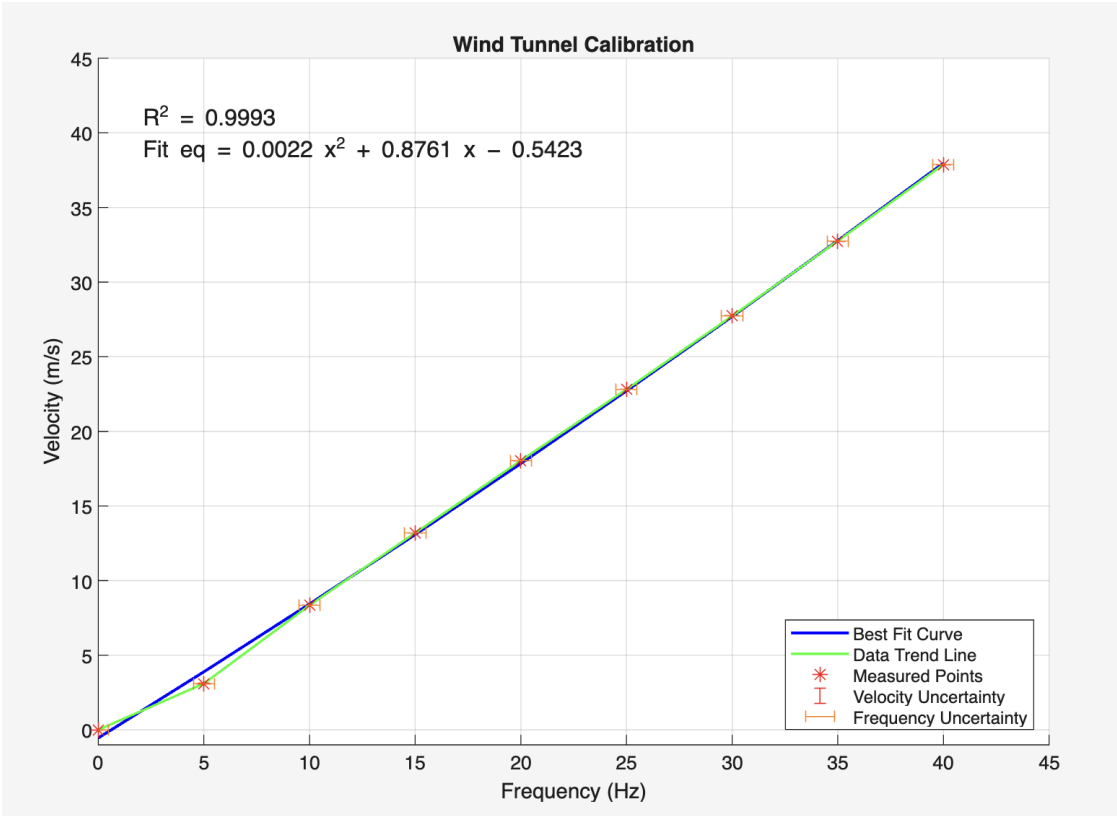

Figure 4: Wind Tunnel Calibration curve with error bars with an equation for the velocity as a function of frequency and coefficient of determination

Table 1: Velocities at different frequencies from wind tunnel calibration and uncertainties

Frequency (Hz)	Velocity (m/s)	Uncertainty - Velocity (m/s)	Uncertainty - Frequency (Hz)
0.0	0.00	± 0.000	± 0.500
5.0	3.11	± 0.0175	± 0.500
10.0	8.35	± 0.00652	± 0.500
15.0	13.20	± 0.00412	± 0.500
20.0	18.04	± 0.00302	± 0.500
25.0	22.81	± 0.00239	± 0.500
30.0	27.73	± 0.00196	± 0.500
35.0	32.75	± 0.00166	± 0.500
40.0	37.88	± 0.00144	± 0.500
40.0 (extrapolated)	37.97	n/a	± 0.500
45.0 (extrapolated)	43.27	n/a	± 0.500

Vortex Shedding

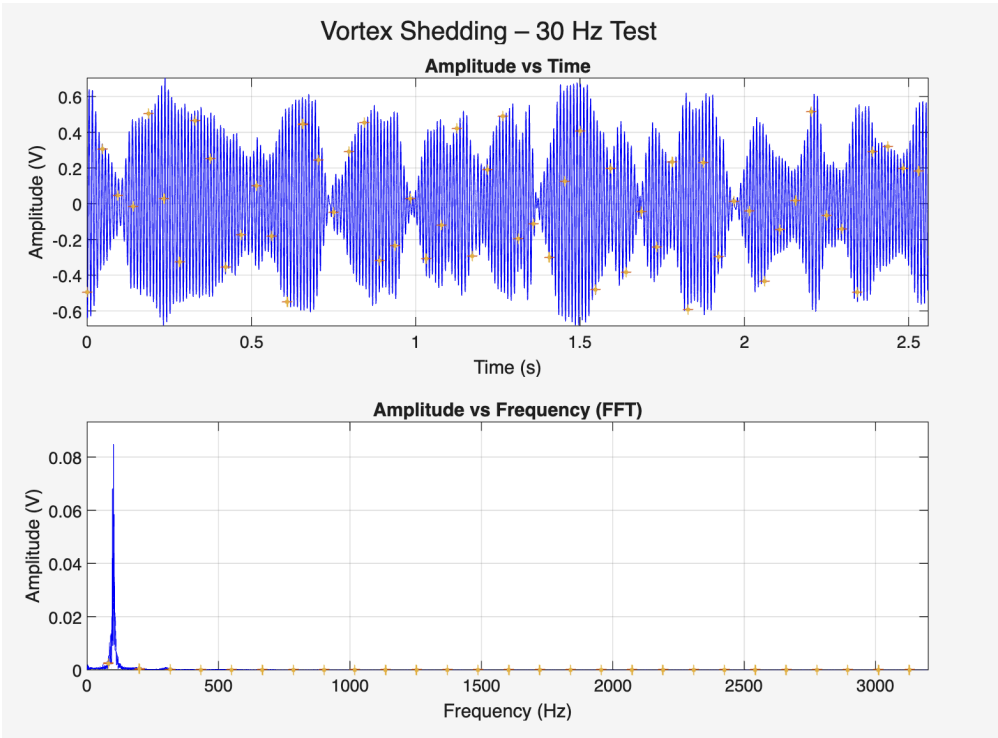

Figure 5: Amplitude vs Time and Amplitude vs Frequency subplots for 30 Hz vortex shedding experiment

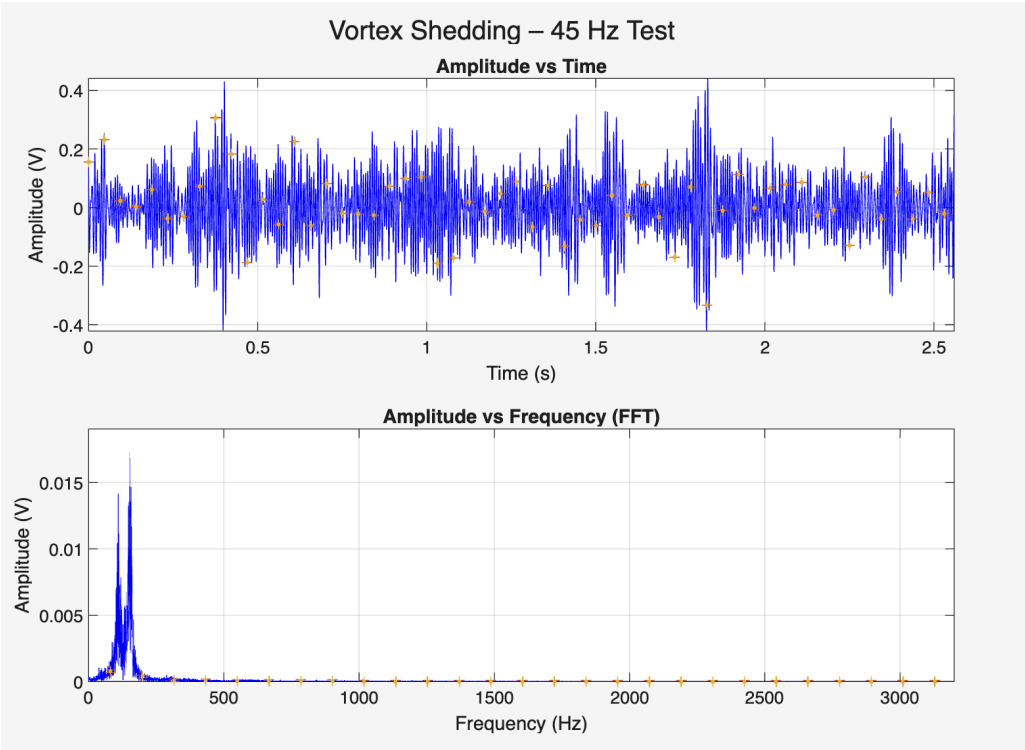

Figure 6: Amplitude vs Time and Amplitude vs Frequency subplots for 30 Hz vortex shedding experiment

Table 2: Frequency, velocity, Strohauls Number, Reynolds number, theoretical and experimental shedding frequencies for both 30 and 45 Hz vortex shedding experiment

	Experiment 1	Uncertainty	Experiment 2	Uncertainty
Tunnel Frequency (Hz)	30.0	± 0.500	45.0	± 0.500
Tunnel Velocity (m/s)	27.727	± 0.00196	43.269	n/a
Strohauls Number	0.198	± 3.595e-07	0.198	± 2.304e-07
Reynolds Number	9.042e+04	± 7.535e+02	1.411e+05	± 1.176e+03
Theoretical Shedding Frequency (Hz)	115.249	± 0.960	179.865	± 1.499
Experimental Shedding Frequency (Hz)	100.391	± 0.500	151.172	± 0.500

Cylinder Wake

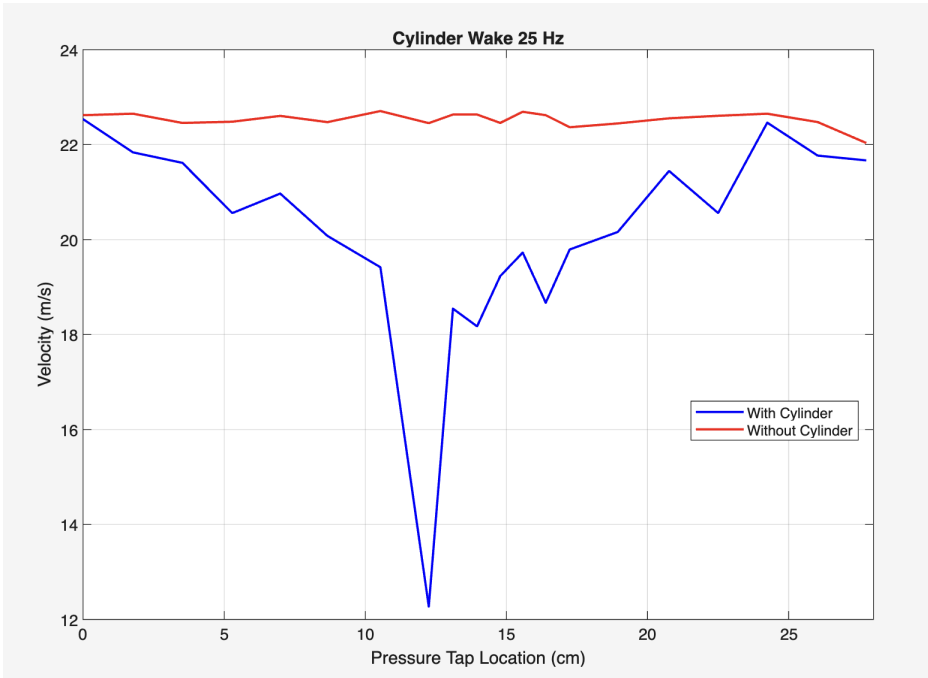

Figure 7: Velocity profile across the rake at 25 Hz fan speed, comparing freestream flow (without cylinder) to wake flow (with cylinder)

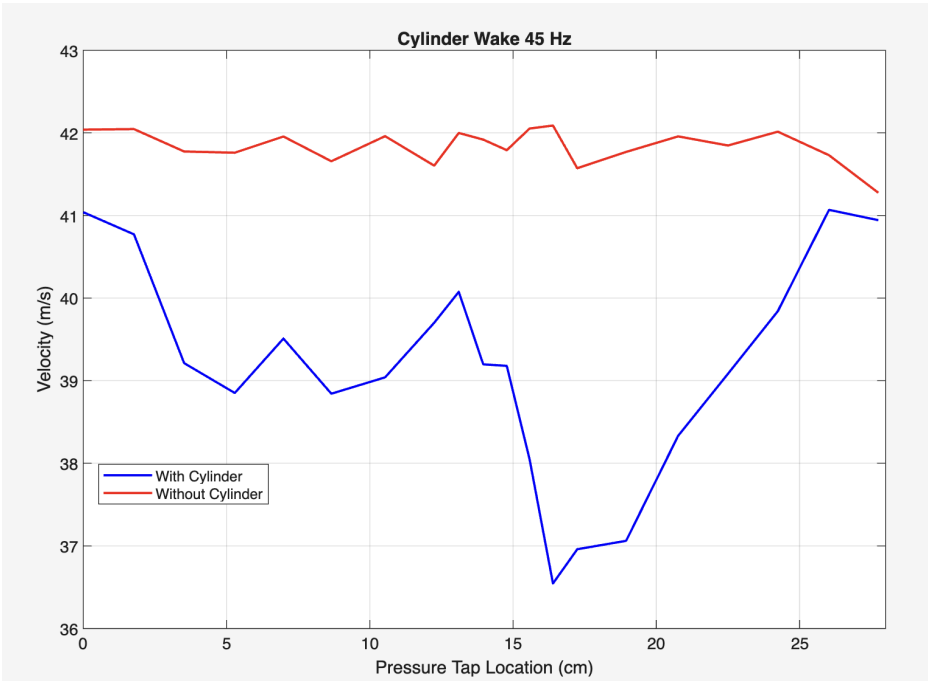

Figure 8: Velocity profile across the rake at 45 Hz fan speed, comparing freestream flow (without cylinder) to wake flow (with cylinder)

Table 3: Cylinder wake drag measurements and coefficients

	25 Hz	Uncertainty	45 Hz	Uncertainty
Tunnel Frequency (Hz)	25	± 0.500	45	± 0.500
Drag Force (N)	8.554	± 0.795	21.409	± 1.073
Drag Coefficient	0.924	± 0.128	0.670	± 0.077
Reynolds number	7.519e+04	± 7.7145e+03	1.396e+05	± 1.433e+04

Drag Derivation

$$-\int\int_{CA} P\,dS \;-\int\int_{CA} P\,dS \;=\; R'$$

$$\frac{\delta}{\delta t}\int\int\int_{V} \rho v dV \;+\int\int_{S}(\rho v \cdot dS)v \;=\; -\int\int_{CA} P \cdot dS \;-\; R'$$

$$\frac{\delta}{\delta t} \;=\; 0$$

$$R' \;=\; -\int\int_{S}(\rho v \cdot dS)v \;-\int\int_{CA} P \cdot dS$$

$$D' \;=\; -\int\int_{S}(\rho v \cdot dS)u \;-\int\int_{CA}(P \cdot dS)_x$$

$$\int\int_{CA}(P \cdot dS)_x \;=\; 0$$

$$D' \;=\; -\int\int_{S}(\rho v \cdot dS)u$$

$$\int\int_{S}(\rho v \cdot dS)u \;=\; -\int_{1} \rho_1 u_1^2 dy \;+\int_{2} \rho_2 u_2^2 dy$$

$$\int\int_{S}(\rho v \cdot dS) \;=\; -\int_{1} \rho_1 u_1 dy \;+\int_{2} \rho_2 u_2 dy \;=\; 0$$

$$D' \;=\int_{2} \rho_2 u_2 (u_1 - u_2) dy$$

$$C_D \;=\; \frac{D}{\frac{1}{2}\rho U_\infty^2 S}$$

$$S \;=\; (Diameter \times Length)$$

Boundary Layer

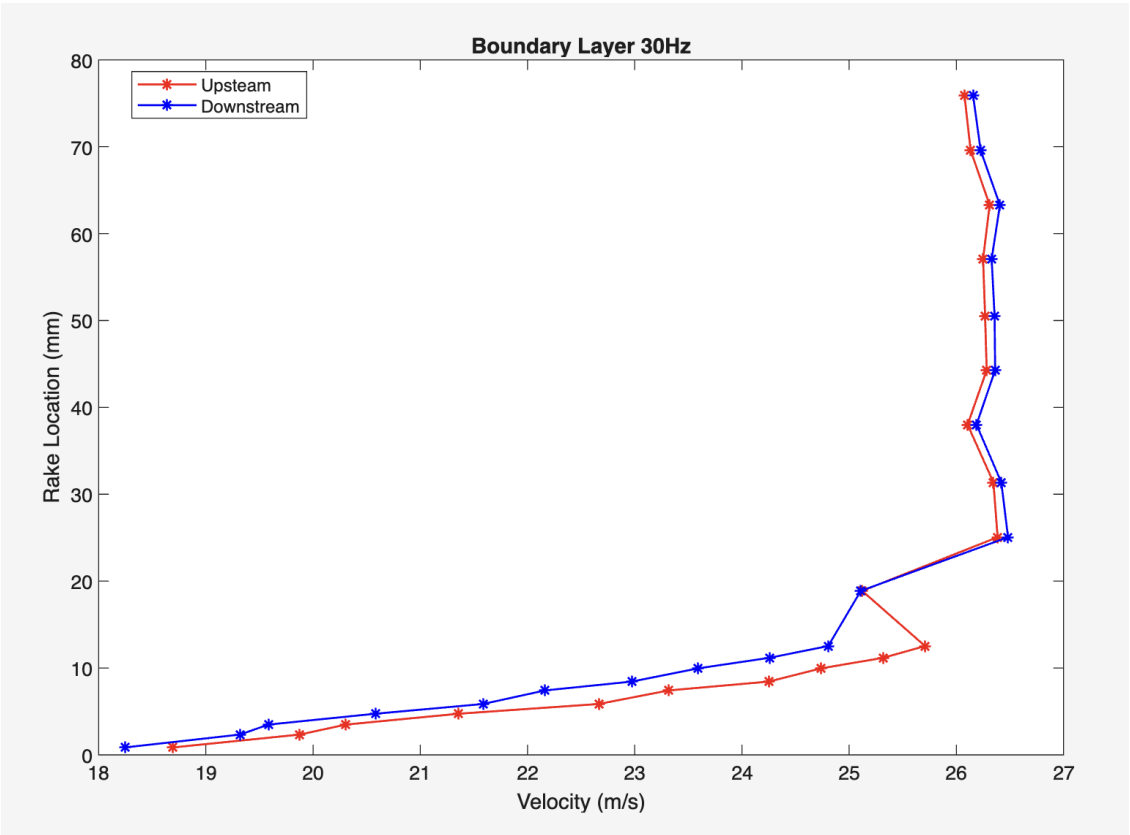

Figure 10: Velocity profile vs distance from wall at 30 Hz, comparing upstream and downstream measurement positions.

Airfoil

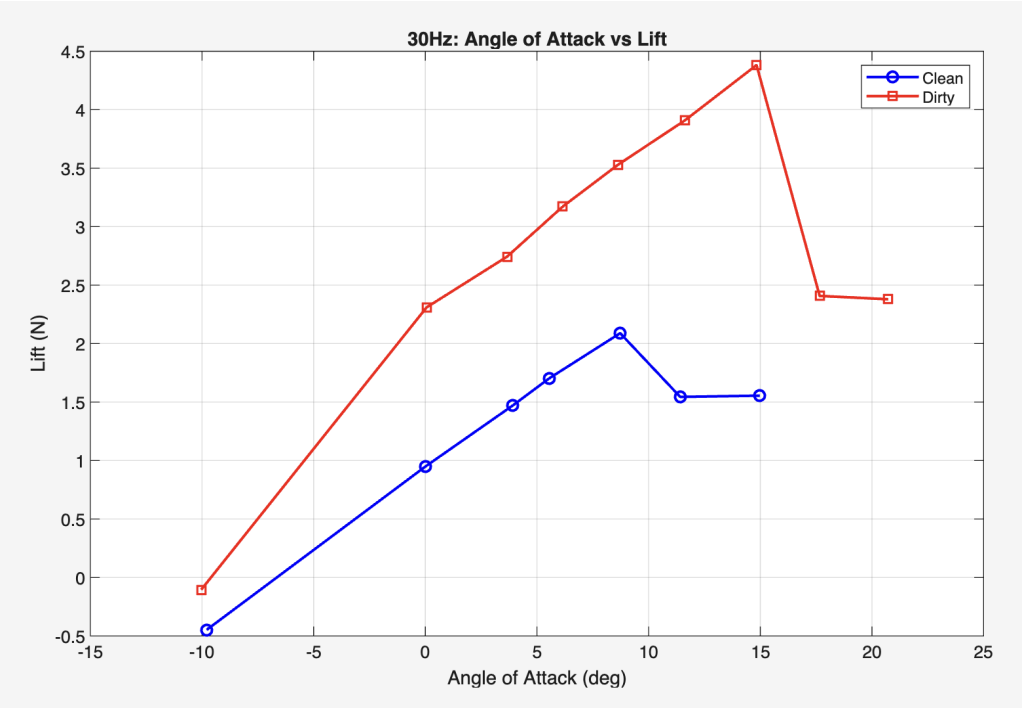

Figure 11: Angle of attack vs Lift at 30 Hz clean and dirty configurations

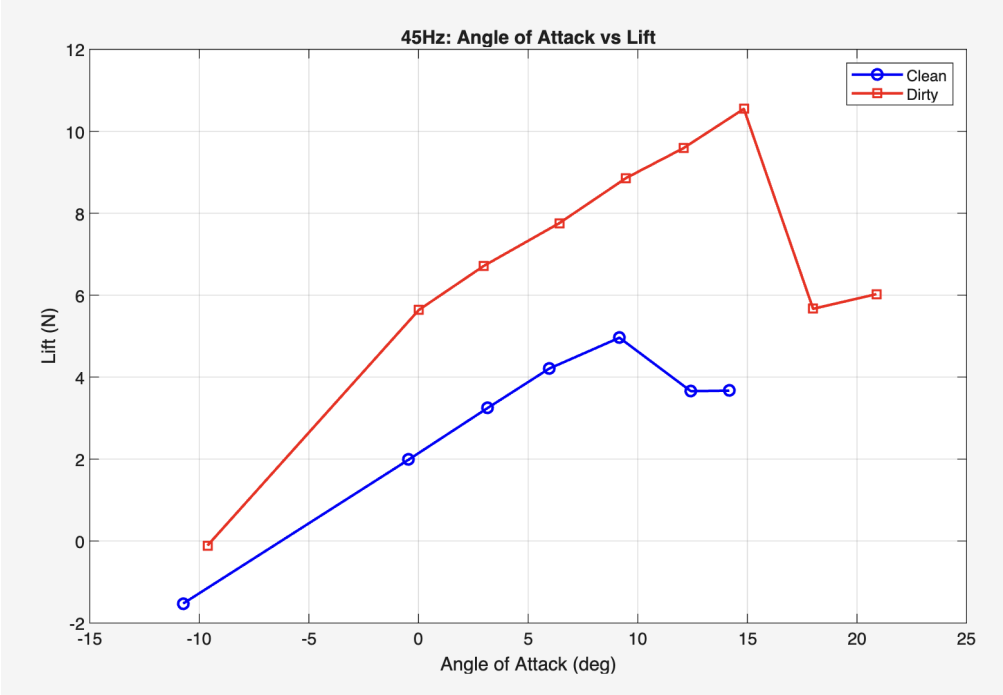

Figure 12: Angle of attack vs Lift at 45 Hz clean and dirty configurations

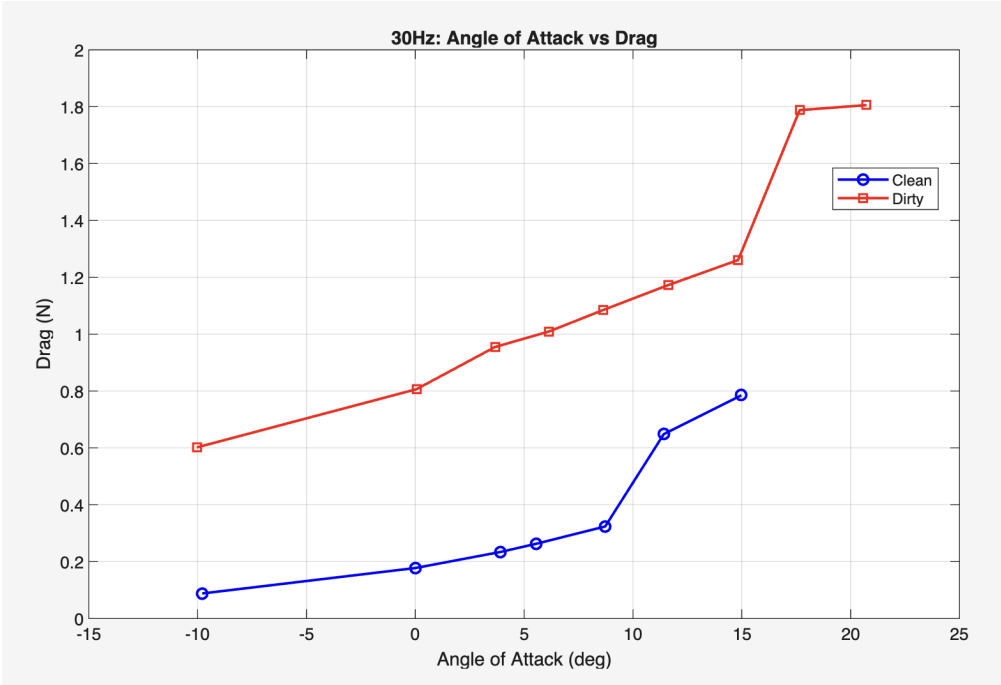

Figure 13: Angle of attack vs Drag at 30 Hz clean and dirty configurations

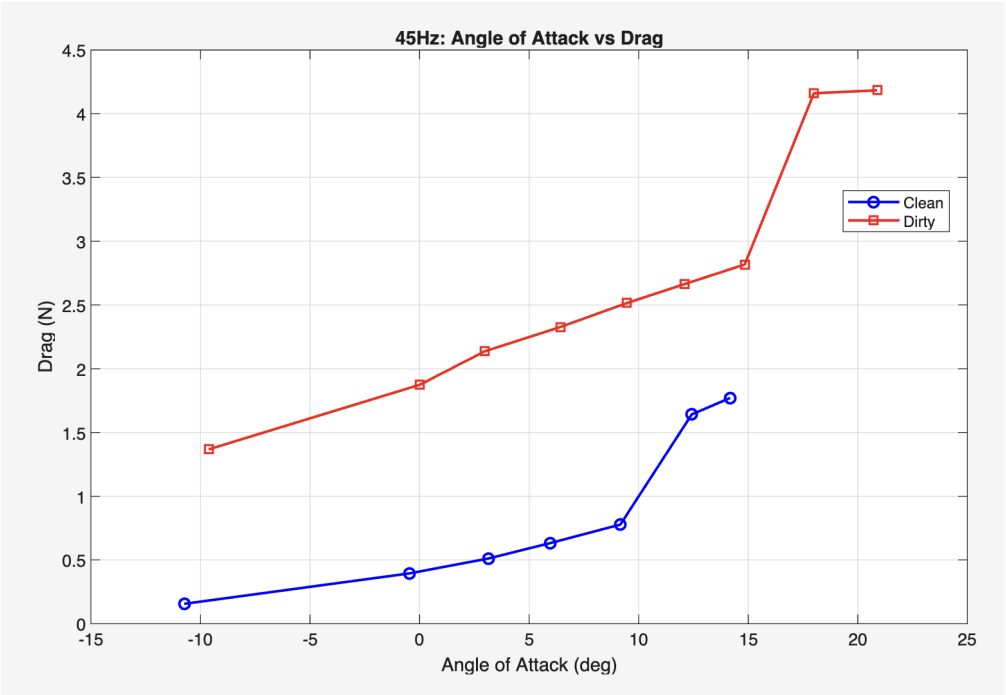

Figure 14: Angle of attack vs Drag at 45 Hz clean and dirty configurations

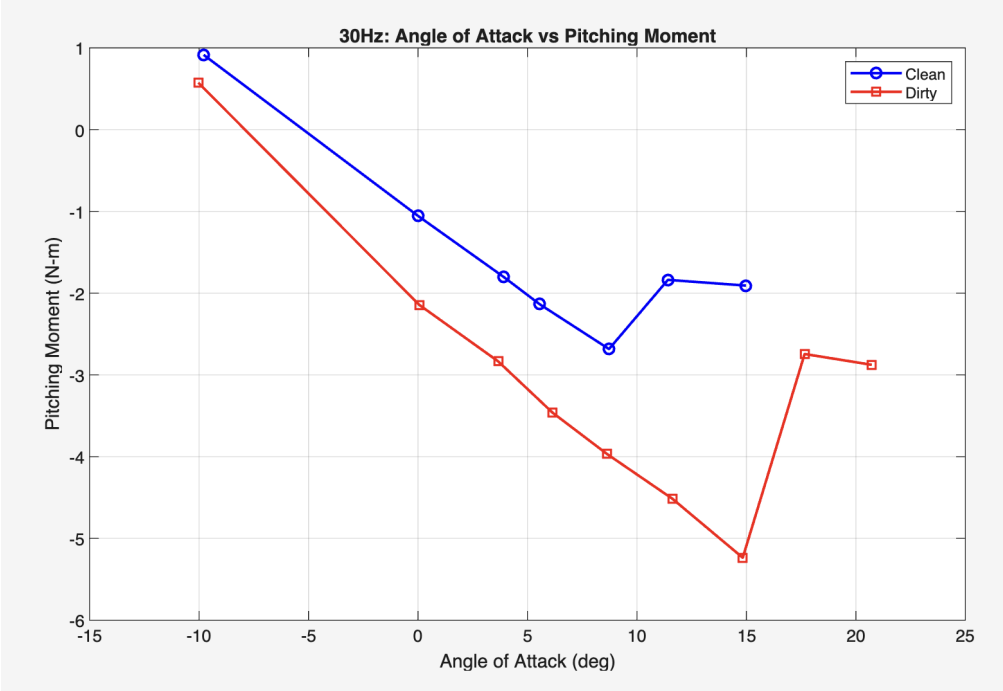

Figure 15: Angle of attack vs Pitching Moment at 30 Hz clean and dirty configurations

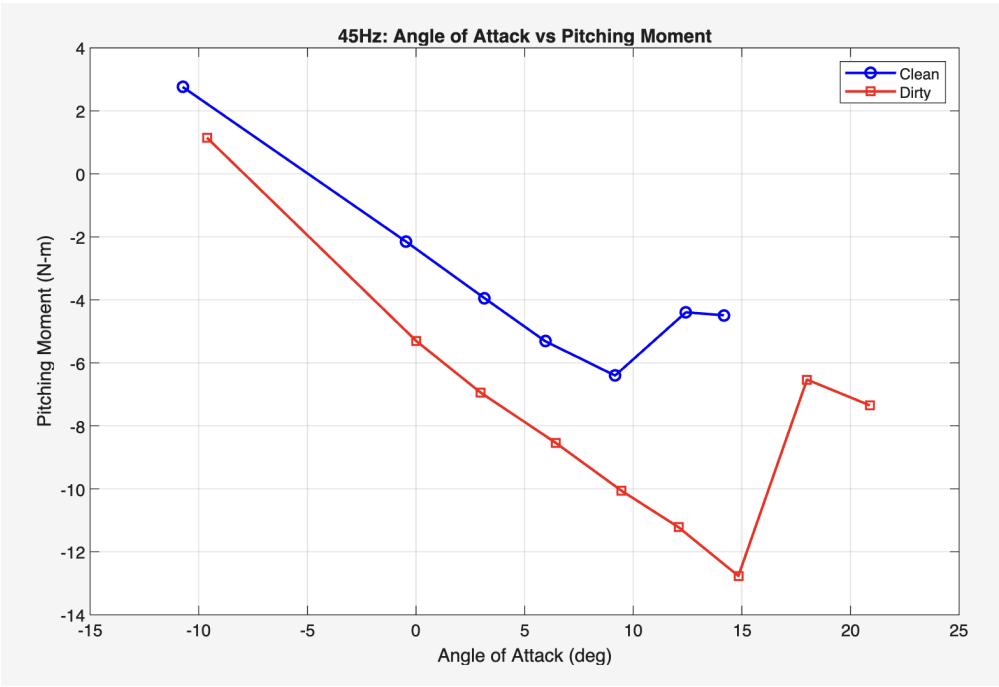

Figure 16: Angle of attack vs Pitching Moment at 45 Hz clean and dirty configurations

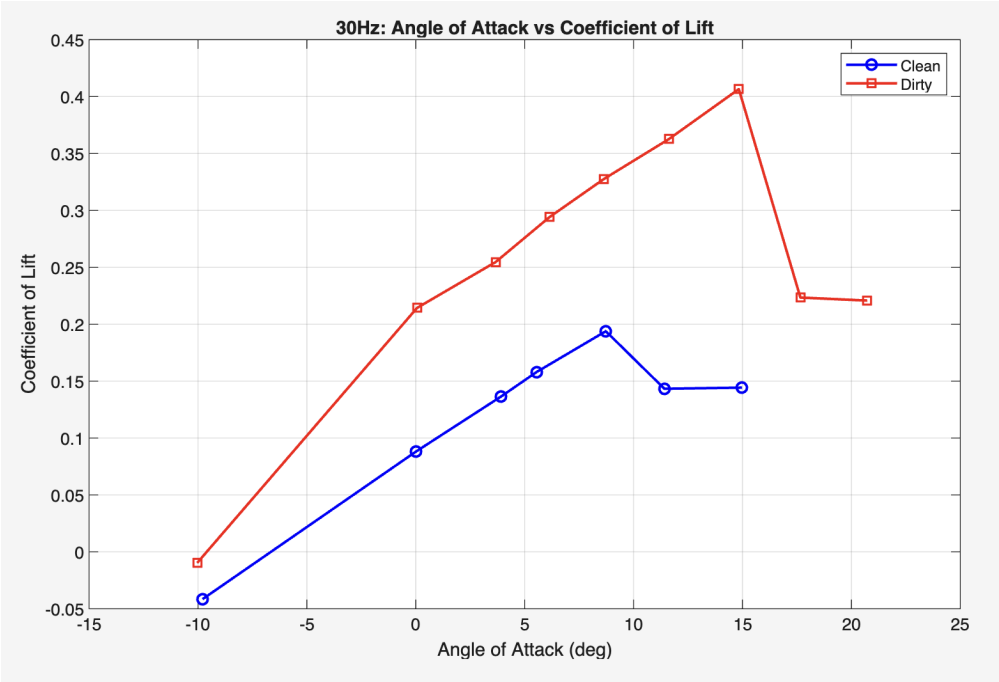

Figure 17: Angle of attack vs Coefficient of Lift at 30 Hz clean and dirty configurations

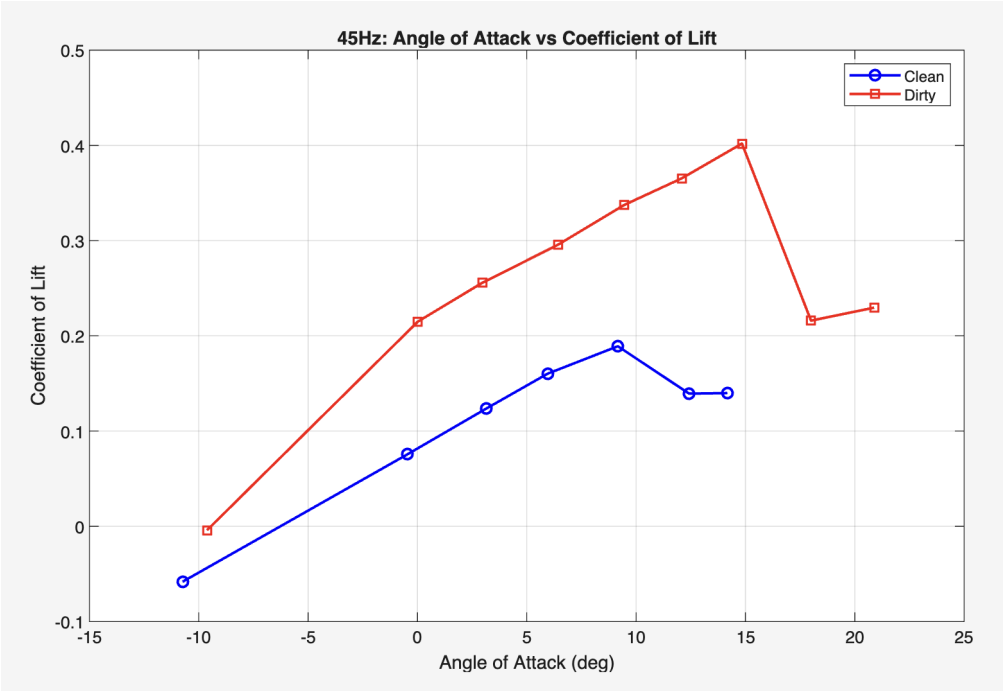

Figure 18: Angle of attack vs Coefficient of Lift at 45 Hz clean and dirty configurations

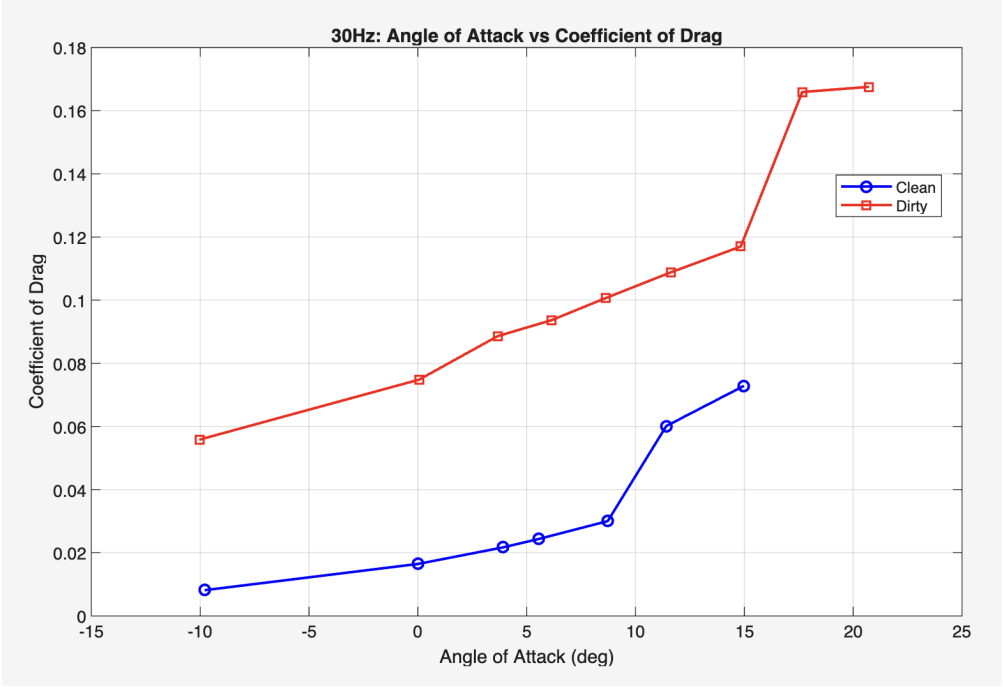

Figure 19: Angle of attack vs Coefficient of Drag at 30 Hz clean and dirty configurations

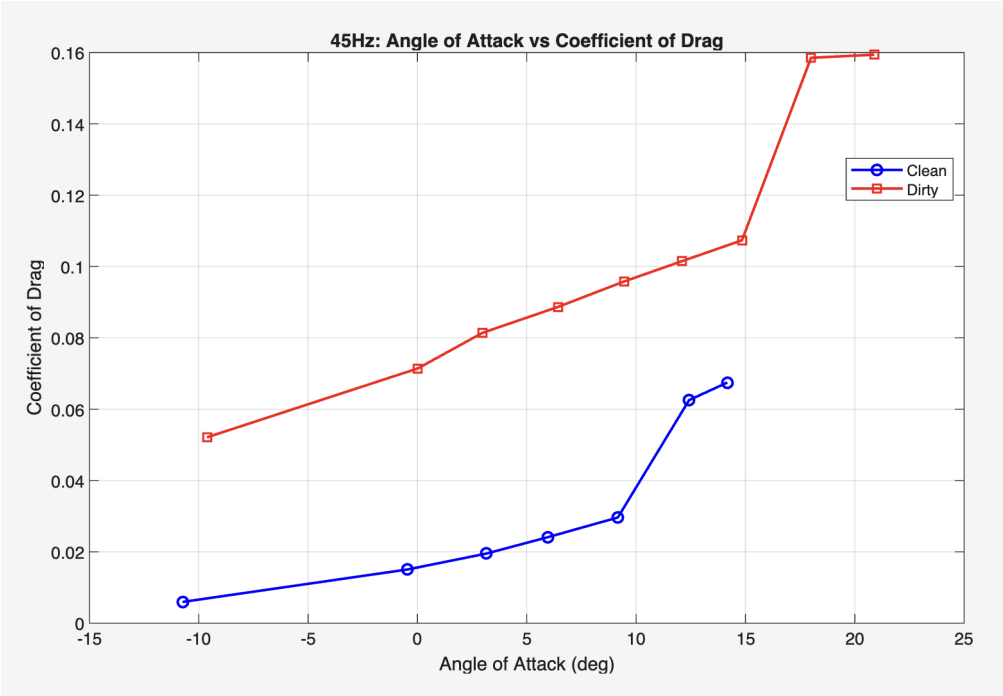

Figure 20: Angle of attack vs Coefficient of Drag at 45 Hz clean and dirty configurations

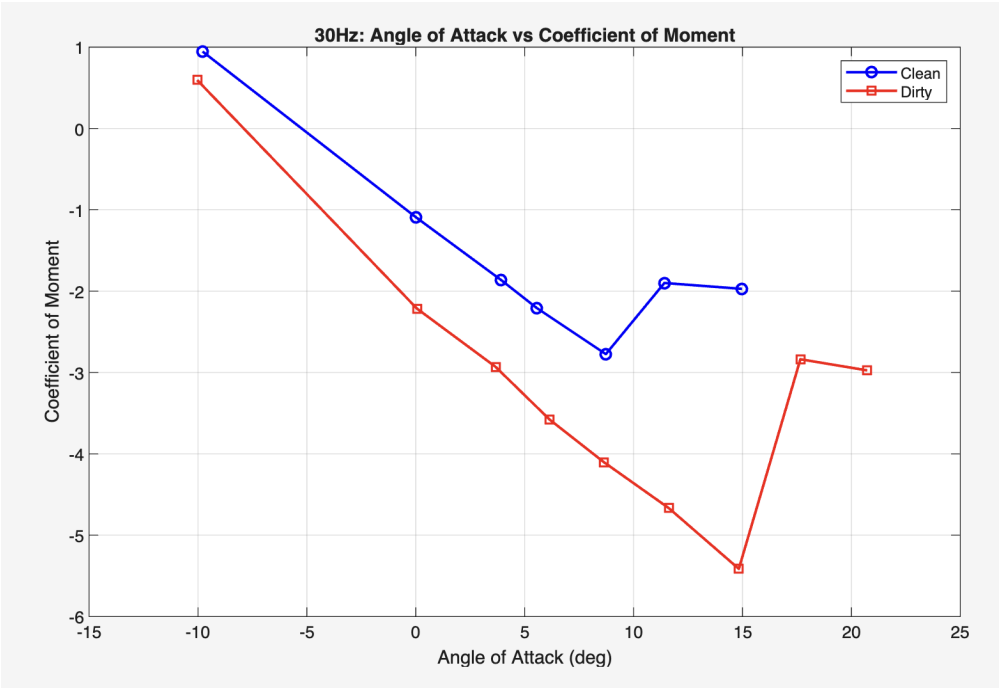

Figure 21: Angle of attack vs Coefficient of Pitching Moment at 30 Hz clean and dirty configurations

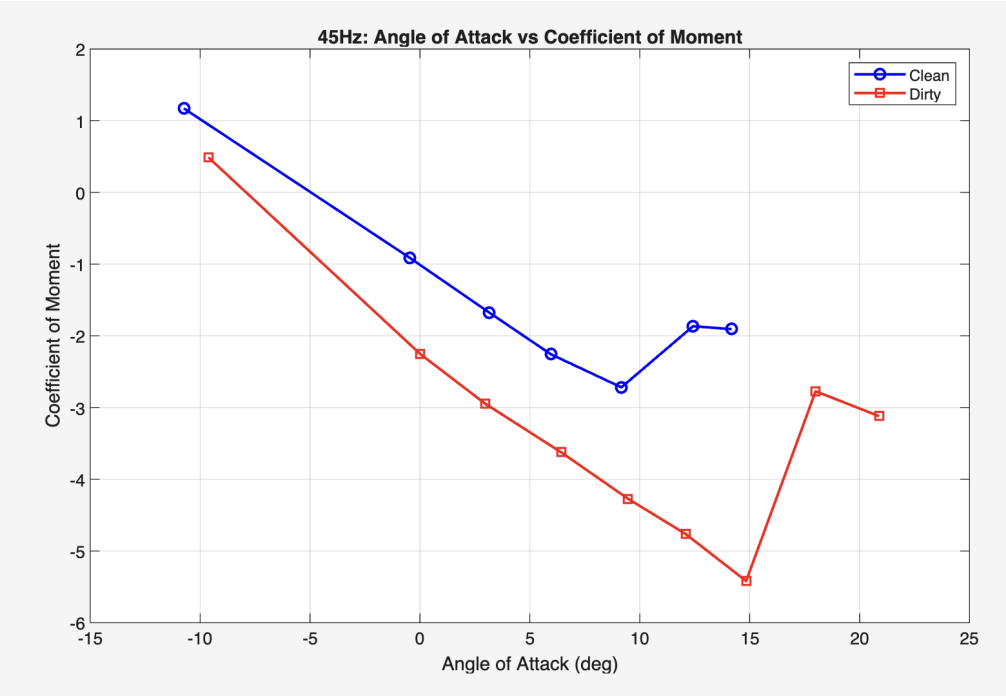

Figure 22: Angle of attack vs Coefficient of Pitching Moment at 45 Hz clean and dirty configurations

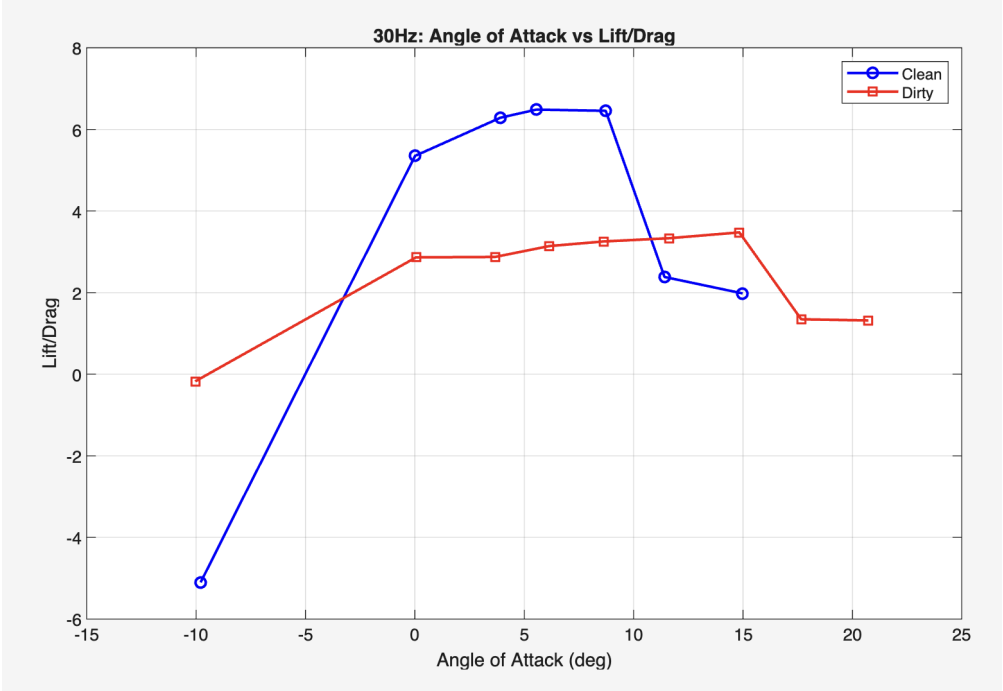

Figure 23: Angle of attack vs Lift-to-Drag ratio at 30 Hz clean and dirty configurations

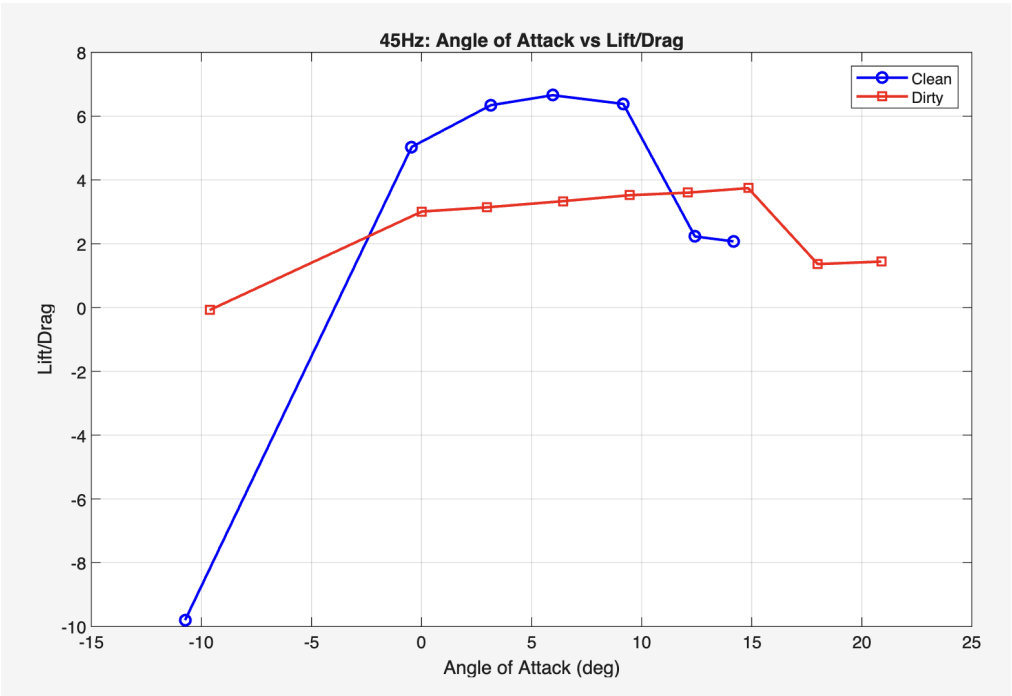

Figure 24: Angle of attack vs Lift-to-Drag ratio at 45 Hz clean and dirty configurations

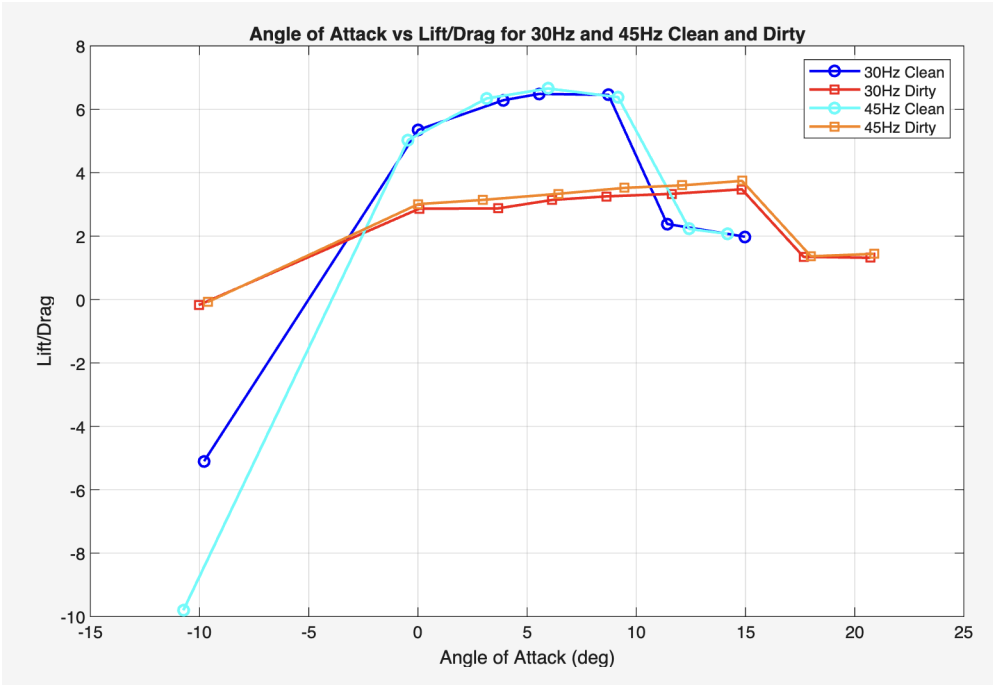

Figure 25: Angle of attack vs Lift-to-Drag ratio for all test conditions (30 Hz and 45 Hz, clean and dirty)

Table 4: Airfoil Aerodynamic Performance Parameters at 30 Hz and 45 Hz, Clean and Dirty

	30 Hz – Clean	30 Hz – Dirty	45 Hz – Clean	45 Hz – Dirty
Tunnel Frequency (Hz)	30.0	30.0	45.0	45.0
Tunnel Velocity (m/s)	27.727	27.727	43.269	43.269
Reynolds Number	1.684e+5	1.684e+5	2.628e+5	2.628e+5
Peak Efficiency (L/D) – Clean	6.49	–	6.66	–
Peak Efficiency (L/D) – Dirty	–	3.48	–	3.74
Lift-Curve Slope (per deg)	0.0127	–	0.0124	–
Lift-Curve Slope (per rad)	0.728	–	0.712	–
Stall Angle (deg)	~9°	~15°	~9°	~15°

2D/3D Cylinder

Table 5: Length, Reynolds number, Drag, and Coefficient of Drag with associated uncertainties for each cylinder

	Cylinder 0 – No cyl	Cylinder 1 – Long	Cylinder 2 – Medium	Cylinder 3 – Short
Length (m)	0.000	0.607	0.507	0.267
Reynolds Number	n/a	9.039e+4	9.077e+4	9.004e+4
Uncertainty in Reynolds Number	n/a	± 11.35	± 11.37	± 11.34
Drag (N)	0.002072	3.746	2.995	1.473
Uncertainty in Drag (N)	± 4.143e-05	± 0.07492	± 0.05991	± 0.02947
Coefficient of Drag	n/a	0.2721	0.2594	0.2442
Uncertainty in Coefficient of Drag	n/a	± 0.00544	± 0.00519	± 0.00488

6.3 Code

```
% Matthew Pepper
% MAE 315
% Lab 5a
clear
clc
close all
%% Wind Tunnel Calibration
fprintf('Experiment 1: \n\n');
% Length between taps (mm) = Length measured (mm) - 1 diameter (mm)
tap_d = 1.62; % all in mm
tap12   = 19.33 - tap_d;
tap23   = 19.14 - tap_d;
tap34   = 19.28 - tap_d;
tap45   = 18.60 - tap_d;
tap56   = 18.34 - tap_d;
tap67   = 20.38 - tap_d;
tap78   = 18.76 - tap_d;
tap89   = 10.19 - tap_d;
tap910  = 10.16 - tap_d;
tap1011 = 9.83  - tap_d;
tap1112 = 9.58  - tap_d;
tap1213 = 9.78  - tap_d;
tap1314 = 10.10 - tap_d;
tap1415 = 18.66 - tap_d;
tap1516 = 19.78 - tap_d;
tap1617 = 19.01 - tap_d;
tap1718 = 19.02 - tap_d;
tap1819 = 19.48 - tap_d;
tap1920 = 18.82 - tap_d;
tap2021 = 20.25 - tap_d;
freq_fan = 0:5:40;        % Hz
tap_dist_mm = [tap12, tap23, tap34, tap45, tap56, tap67, tap78, tap89, ...
   tap910, tap1011, tap1112, tap1213, tap1314, tap1415, tap1516, ...
   tap1617, tap1718, tap1819, tap1920, tap2021];
tap_dist_m = tap_dist_mm ./ 1000;   % mm to m
% Raw measured pressures (Torr)
p1 = [-0.003  0.044 0.322 0.801 1.496 2.399 3.537 4.925 6.595];
p2 = [-0.001  0.043 0.319 0.797 1.488 2.384 3.533 4.933 6.588];
p3 = [-0.003  0.046 0.320 0.802 1.501 2.390 3.526 4.927 6.592];
% Construct vectors for each frequency
P_f0  = [p1(1)  p2(1)  p3(1)];
P_f5  = [p1(2)  p2(2)  p3(2)];
P_f10 = [p1(3)  p2(3)  p3(3)];
P_f15 = [p1(4)  p2(4)  p3(4)];
P_f20 = [p1(5)  p2(5)  p3(5)];
P_f25 = [p1(6)  p2(6)  p3(6)];
P_f30 = [p1(7)  p2(7)  p3(7)];
P_f35 = [p1(8)  p2(8)  p3(8)];
P_f40 = [p1(9)  p2(9)  p3(9)];
% Create averaged scalar values
P_f0avg  = mean(P_f0);
P_f5avg  = mean(P_f5);
P_f10avg = mean(P_f10);
P_f15avg = mean(P_f15);
```

```
P_f20avg = mean(P_f20);
P_f25avg = mean(P_f25);
P_f30avg = mean(P_f30);
P_f35avg = mean(P_f35);
P_f40avg = mean(P_f40);
dyn_Ptorr = [P_f0avg, P_f5avg, P_f10avg, P_f15avg, P_f20avg, P_f25avg, ...
   P_f30avg, P_f35avg, P_f40avg];
dyn_P = dyn_Ptorr .* 133.322;  % 1 Torr = 133.322 Pa
% Replace small negative pressures with zero
dyn_P(dyn_P < 0) = 0;
% Air density
rho = 1.225; % kg/m^3
% Velocity calculation
vel_air = sqrt(2 .* dyn_P ./ rho);
% Uncertainty
u_P = 0.5 * 0.001 * 133.322; % Pa
% Avoid division by zero for first point
P_for_uncertainty = dyn_P;
P_for_uncertainty(P_for_uncertainty == 0) = 1e-6; % small number for
calculation
u_vel = u_P ./ sqrt(2 .* rho .* P_for_uncertainty);
u_vel(dyn_P == 0) = 0; % set first point uncertainty to 0
% Frequency uncertainty
u_freq = 0.5; % Hz
eb_freq = u_freq * ones(size(freq_fan));
eb_vel = u_vel;
% Quadratic Fit
p = polyfit(freq_fan, vel_air, 2); % v = a*f^2 + b*f + c
vel_fit = polyval(p, freq_fan);
% R-squared
R = corrcoef(vel_air, vel_fit);
R_squared = R(1,2)^2;
fit_eq_str = sprintf('Fit eq = %.4f x^2 + %.4f x - %.4f', p(1), p(2), ...
   -p(3));
% Plot calibration curve with uncertainties
figure(1)
hold on
% Fit line
h_fit = plot(freq_fan, vel_fit, 'b', 'LineWidth', 1.5);
% Line of points
h_data_line = plot(freq_fan, vel_air, 'g-', 'LineWidth', 1.2);
% Measured points
h_meas = plot(freq_fan, vel_air, 'r*', 'MarkerSize', 8);
% Vertical error bars (velocity)
h_vert_err = errorbar(freq_fan, vel_air, eb_vel, 'vertical', ...
   'LineStyle', 'none', 'Color', [1 0 0]);
% Horizontal error bars (frequency)
h_horiz_err = errorbar(freq_fan, vel_air, eb_freq, 'horizontal', ...
   'LineStyle', 'none', 'Color', [1 0.5 0]);
xlabel('Frequency (Hz)')
ylabel('Velocity (m/s)')
title('Wind Tunnel Calibration')
% Updated legend including error bars
legend([h_fit, h_data_line, h_meas, h_vert_err, h_horiz_err], ...
      {'Best Fit Curve', 'Data Trend Line', 'Measured Points', ...
       'Velocity Uncertainty', 'Frequency Uncertainty'}, ...
      'Location', 'southeast')
```

```
% Display fit equation and R-squared
text(2, 40, {['R^2 = ', num2str(R_squared, '%.4f')], fit_eq_str}, ...
   'FontSize', 12)
grid on;
xlim([0 45]);
ylim([-1 45])
hold off
% Estimate velocities
vel_40Hz = polyval(p, 40);
vel_45Hz = polyval(p, 45);
fprintf('Velocity at 40 Hz: %.3f m/s\n', vel_40Hz);
fprintf('Velocity at 45 Hz: %.3f m/s\n', vel_45Hz);
fprintf('\n')
% Tabulate results including estimated points
freq_table = [freq_fan, 40, 45];             % measured + estimated frequencies
vel_table  = [vel_air, vel_40Hz, vel_45Hz]; % measured + estimated velocities
u_vel_table = [u_vel, NaN, NaN];            % measured uncertainties, NaN for
estimates
u_freq_table = u_freq * ones(size(freq_table));
T = table(freq_table', vel_table', u_vel_table', u_freq_table', ...
   'VariableNames', {'Frequency_Hz', 'Velocity_m_per_s', ...
   'Velocity_Uncertainty', 'Frequency_Uncertainty'});
disp(T)
%% Vortex Shedding - 30 Hz
fprintf('\n');
fprintf('Experiment 2: \n');
% 30 Hz
%  Uncertainties
uV  = 0.0001;        % Voltage uncertainty (V)
uF  = 0.5;           % Frequency uncertainty (Hz)
uT  = 1/6400;        % Time uncertainty (s)
%  Import 30 Hz data
opts = detectImportOptions('vortex_30.txt');
opts.Delimiter = {' ',','};
Data = readtable('vortex_30.txt', opts);
t30   = str2double(Data.t);
amp30 = str2double(Data.ExtraVar1);
% error bar vectors
err_t = uT * ones(size(t30));
err_A = uV * ones(size(amp30));
% time domain
figure(2)
sgtitle('Vortex Shedding - 30 Hz Test')
subplot(2,1,1)
plot(t30, amp30, 'b')
hold on
% error bars
idx = 1:300:length(t30);
errorbar(t30(idx), amp30(idx), err_A(idx), '.', 'vertical');
errorbar(t30(idx), amp30(idx), err_t(idx), '.', 'horizontal');
xlabel('Time (s)')
ylabel('Amplitude (V)')
title('Amplitude vs Time')
xlim([0 max(t30)])
ylim([min(amp30) max(amp30)])
grid on
hold off
```

```
% fft
Fs = 6400;                % sampling freq
N  = length(amp30);
df = Fs/N;
Y = fftshift(fft(amp30));
f = -Fs/2 : df : Fs/2-df;
ampFFT = abs(Y)/N;
% Uncertainties
err_F = uF * ones(size(f));
err_Afft = uV * ones(size(ampFFT));
subplot(2,1,2)
plot(f, ampFFT, 'b')
hold on
idx = 1:300:length(f);
errorbar(f(idx), ampFFT(idx), err_Afft(idx), '.', 'vertical');
errorbar(f(idx), ampFFT(idx), err_F(idx), '.',   'horizontal');
xlabel('Frequency (Hz)')
ylabel('Amplitude (V)')
title('Amplitude vs Frequency (FFT)')
xlim([0 Fs/2])
ylim([0 max(ampFFT)*1.1])
grid on
hold off
%  30 Hz Tunnel Velocity and Uncertainty
v30 = 27.727;          % m/s from wind tunnel calibration
uV30 = 0.0019626;      % m/s
%  Reynolds Number and Uncertainty
rho = 1.225;          % kg/m^3
visc = 0.00001789;    % viscosity kg/(m·s)
D = 1.875 * 0.0254;   % cylinder diameter (m)
uD = (1/32)*0.0254*0.5;  % diameter uncertainty
uV = uV30;
Re30 = rho * v30 * D / visc;
% Uncertainty propagation:
uRe30 = Re30 * sqrt( (uV/v30)^2 + (uD/D)^2 );
%  Strouhal Number and Uncertainty (empirical curve)
St30 = 0.198 * (1 - 19.7/Re30);
% uncertainty from derivative
dSt_dRe = 0.198 * (19.7 / Re30^2);
uSt30 = abs(dSt_dRe) * uRe30;
%  Empirical Shedding Frequency and Uncertainty (propagate)
shf30 = St30 * v30 / D;
% df/dRe = ( (v/D) * dSt/dRe )
term1 = (v30 / D) * dSt_dRe * uRe30;
% df/dv = (St/D)
term2 = (St30 / D) * uV;
% df/dD = -(St*v / D^2)
term3 = abs(St30 * v30 / D^2) * uD;
ushf30 = sqrt(term1^2 + term2^2 + term3^2);
%  Display 30 Hz Results in scientific notation for small uncertainties
fprintf('\n30 Hz Vortex Shedding Results\n')
fprintf('Tunnel Frequency:       30 ± %.3f Hz\n', uF)
fprintf('Tunnel Velocity:        %.3f ± %.6f m/s\n', v30, uV30)
fprintf('Reynolds Number:        %.3e ± %.4e\n', Re30, uRe30)
fprintf('Strouhal Number:        %.6f ± %.4e\n', St30, uSt30)
fprintf('Empirical Shed. Freq:   %.4f ± %.4f Hz\n', shf30, ushf30)
%% Vortex Shedding - 45 Hz
```

```
%  Uncertainties
uV = 0.0001;        % Voltage uncertainty (V)
uF = 0.5;           % Frequency uncertainty (Hz)
u = 1/6400;         % Time uncertainty (s)
%  Import 45 Hz data
opts = detectImportOptions('vortex_45.txt');
opts.Delimiter = {' ',','};
Data = readtable('vortex_45.txt', opts);
t45 = str2double(Data.t);
amp45 = str2double(Data.ExtraVar1);
% error bar vectors
err_t = uT * ones(size(t45));
err_A = uV * ones(size(amp45));
% time domain
figure(3)
sgtitle('Vortex Shedding - 45 Hz Test')
subplot(2,1,1)
plot(t45, amp45, 'b')
hold on
% error bars
idx = 1:300:length(t45);
errorbar(t45(idx), amp45(idx), err_A(idx), '.', 'vertical');
errorbar(t45(idx), amp45(idx), err_t(idx), '.', 'horizontal');
xlabel('Time (s)')
ylabel('Amplitude (V)')
title('Amplitude vs Time')
xlim([0 max(t45)])
ylim([min(amp45) max(amp45)])
grid on
hold off
% fft
Fs = 6400;                % sampling freq
N = length(amp45);
df = Fs/N;
Y = fftshift(fft(amp45));
f = -Fs/2 : df : Fs/2-df;
ampFFT = abs(Y)/N;
% Uncertainties
err_F = uF * ones(size(f));
err_Afft = uV * ones(size(ampFFT));
subplot(2,1,2)
plot(f, ampFFT, 'b')
hold on
idx = 1:300:length(f);
errorbar(f(idx), ampFFT(idx), err_Afft(idx), '.', 'vertical');
errorbar(f(idx), ampFFT(idx), err_F(idx), '.',   'horizontal');
xlabel('Frequency (Hz)')
ylabel('Amplitude (V)')
title('Amplitude vs Frequency (FFT)')
xlim([0 Fs/2])
ylim([0 max(ampFFT)*1.1])
grid on
hold off
%  45 Hz Tunnel Velocity (from calibration)
v45 = 43.269;         % m/s, and uV45 is unknown (extrapolated from part 1)
%  Reynolds Number
rho = 1.225;              % kg/m^3
```

```
visc = 0.00001789;        % dynamic viscosity kg/(m·s)
D = 1.875 * 0.0254;       % cylinder diameter (m)
uD = (1/32)*0.0254*0.5;   % diameter uncertainty
% Reynolds number
Re45 = rho * v45 * D / visc;
% Only diameter contributes to uncertainty (velocity uncertainty unknown)
uRe45 = Re45 * (uD / D);
% Strouhal number (empirical)
St45 = 0.198 * (1 - 19.7 / Re45);
% Uncertainty in Strouhal number comes only from Re uncertainty
dSt_dRe = 0.198 * (19.7 / Re45^2);
uSt45   = abs(dSt_dRe) * uRe45;
% Empirical shedding frequency
shf45 = St45 * v45 / D;
% Uncertainty propagation for shedding frequency
term1 = (v45 / D) * dSt_dRe * uRe45;  % from Re
term2 = 0;                            % velocity uncertainty unknown
term3 = abs(St45 * v45 / D^2) * uD;  % from diameter
ushf45 = sqrt(term1^2 + term2^2 + term3^2);
%  Display 45 Hz Results
fprintf('\n45 Hz Vortex Shedding Results\n')
fprintf('Tunnel Frequency:       45 ± %.3f Hz\n', uF)
fprintf('Tunnel Velocity:        %.3f m/s ± NA\n', v45)
fprintf('Reynolds Number:        %.4e ± %.4e\n', Re45, uRe45)
fprintf('Strouhal Number:        %.6f ± %.4e\n', St45, uSt45)
fprintf('Empirical Shed. Freq:   %.4f ± %.4f Hz\n', shf45, ushf45)
%% Cylinder Wake - 25 Hz
fprintf('\n');
fprintf('\n');
fprintf('Experiment 3: \n');
rho = 1.225;          % kg/m^3
psi_to_Pa = 6894.76;  % psi to Pa
% Load data
raw_cyl25 = readmatrix('cyl_25Hz');
raw_cyl25 = raw_cyl25(:,1:20);
cyl25_data = raw_cyl25(:,1:20);
cyl25_avg = mean(cyl25_data,1);
cyl25_avg = cyl25_avg * psi_to_Pa;
cyl25_velocity = sqrt(2 * cyl25_avg / rho);
raw_nocyl25 = readmatrix('nocyl_25Hz');
raw_nocyl25 = raw_nocyl25(:,1:20);
nocyl25_data = raw_nocyl25(:,1:20);
nocyl25_avg = mean(nocyl25_data,1);
nocyl25_avg = nocyl25_avg * psi_to_Pa;
nocyl25_velocity = sqrt(2 * nocyl25_avg / rho);
% Tap positions
tap_dist_cm = tap_dist_mm / 10;
tap_pos_cm = [0, cumsum(tap_dist_cm)];
tap_pos_cm = tap_pos_cm(1:20);
tap_pos_m  = tap_pos_cm / 100;
% Plot velocity profile
figure(4)
plot(tap_pos_cm, cyl25_velocity, 'b-', 'LineWidth', 1.4); hold on;
plot(tap_pos_cm, nocyl25_velocity, 'r-', 'LineWidth', 1.4);
xlabel('Pressure Tap Location (cm)');
ylabel('Velocity (m/s)');
title('Cylinder Wake 25 Hz');
```

```
xlim([0 28])
legend('With Cylinder','Without Cylinder', 'Location', 'best');
grid on;
% Wake Deficit Drag Calculation
U_inf = mean(nocyl25_velocity);   % freestream velocity
uU_inf = std(nocyl25_velocity)/sqrt(length(nocyl25_velocity)); % uVel
u = cyl25_velocity;
du = U_inf - u;
% Drag per unit span
D25_per_m = rho * trapz(tap_pos_m, u .* du);
% Cylinder dimensions
D = 48.74 / 1000;      % diameter (m)
uD = 0.005;             % diameter uncertainty (m)
L = 0.611;              % length (m)
uL = 0.0005;            % length uncertainty (m)
S = D * L;              % projected area
% Total drag
D25_total = D25_per_m * L;
% Uncertainty in drag from velocity variations
u_vel_vector = std(cyl25_velocity)/sqrt(length(cyl25_velocity)) * ...
   ones(size(cyl25_velocity));
dx = diff([0, tap_pos_m]);
dD_du = rho * (U_inf - 2 * cyl25_velocity) .* dx;
dD_dU = rho * trapz(tap_pos_m, cyl25_velocity);
uDrag = sqrt( sum( (dD_du .* u_vel_vector).^2 ) + (dD_dU * uU_inf)^2 );
% Drag Coefficient and Uncertainty
Cd25 = D25_total / (0.5 * rho * U_inf^2 * S);
% Relative uncertainties
rel_num = uDrag / D25_total;
rel_den = sqrt( (2*uU_inf/U_inf)^2 + (uD/D)^2 + (uL/L)^2 );
rel_Cd = sqrt(rel_num^2 + rel_den^2);
uCd25 = Cd25 * rel_Cd;
% Reynolds Number and Uncertainty
mu = 0.00001789;   % dynamic viscosity kg/(m·s)
Re25 = (rho * U_inf * D) / mu;
uRe25 = Re25 * sqrt( (uU_inf/U_inf)^2 + (uD/D)^2 );
% Display Results
fprintf('\n25 Hz Cylinder Wake Results:\n');
fprintf('Drag = %.4f ± %.4f N\n', D25_total, uDrag);
fprintf('Drag Coefficient = %.4f ± %.4f\n', Cd25, uCd25);
fprintf('Reynolds Number = %.4e ± %.4e\n', Re25, uRe25);
%% Cylinder Wake - 45 Hz
% Load data
raw_cyl45 = readmatrix('cyl_45Hz');
raw_cyl45 = raw_cyl45(:,1:20);
cyl45_data = raw_cyl45(:,1:20);
cyl45_avg = mean(cyl45_data,1);
cyl45_avg = cyl45_avg * psi_to_Pa;
cyl45_velocity = sqrt(2 * cyl45_avg / rho);
raw_nocyl45 = readmatrix('nocyl_45Hz');
raw_nocyl45 = raw_nocyl45(:,1:20);
nocyl45_data = raw_nocyl45(:,1:20);
nocyl45_avg = mean(raw_nocyl45,1);
nocyl45_avg = nocyl45_avg * psi_to_Pa;
nocyl45_velocity = sqrt(2 * nocyl45_avg / rho);
% Tap positions
tap_pos_cm = [0, cumsum(tap_dist_mm / 10)];
```

```
tap_pos_cm = tap_pos_cm(1:20);
tap_pos_m  = tap_pos_cm / 100;
% Plot velocity profile
figure(5)
plot(tap_pos_cm, cyl45_velocity, 'b-', 'LineWidth', 1.4); hold on;
plot(tap_pos_cm, nocyl45_velocity, 'r-', 'LineWidth', 1.4);
xlabel('Pressure Tap Location (cm)');
ylabel('Velocity (m/s)');
title('Cylinder Wake 45 Hz');
xlim([0 28])
legend('With Cylinder','Without Cylinder', 'Location', 'best');
grid on;
% Wake Deficit Drag Calculation
U_inf = mean(nocyl45_velocity);   % freestream velocity
uU_inf = std(nocyl45_velocity)/sqrt(length(nocyl45_velocity)); % uVel
u = cyl45_velocity;
du = U_inf - u;
% Drag per unit span
D45_per_m = rho * trapz(tap_pos_m, u .* du);
% Cylinder dimensions
D = 48.74 / 1000;      % diameter (m)
uD = 0.005;             % diameter uncertainty (m)
L = 0.611;              % length (m)
uL = 0.0005;            % length uncertainty (m)
S = D * L;              % projected area
% Total drag
D45_total = D45_per_m * L;
% Uncertainty in drag from velocity variations
u_vel_vector = std(cyl45_velocity)/sqrt(length(cyl45_velocity)) * ...
   ones(size(cyl45_velocity));
dx = diff([0, tap_pos_m]);
dD_du = rho * (U_inf - 2 * cyl45_velocity) .* dx;
dD_dU = rho * trapz(tap_pos_m, cyl45_velocity);
uDrag = sqrt( sum( (dD_du .* u_vel_vector).^2 ) + (dD_dU * uU_inf)^2 );
% Drag Coefficient and Uncertainty
Cd45 = D45_total / (0.5 * rho * U_inf^2 * S);
% Relative uncertainties
rel_num = uDrag / D45_total;
rel_den = sqrt( (2*uU_inf/U_inf)^2 + (uD/D)^2 + (uL/L)^2 );
rel_Cd = sqrt(rel_num^2 + rel_den^2);
uCd45 = Cd45 * rel_Cd;
% Reynolds Number and Uncertainty
mu = 0.00001789;   % dynamic viscosity kg/(m·s)
Re45 = (rho * U_inf * D) / mu;
uRe45 = Re45 * sqrt( (uU_inf/U_inf)^2 + (uD/D)^2 );
% Display Results
fprintf('\n45 Hz Cylinder Wake Results:\n');
fprintf('Drag = %.4f ± %.4f N\n', D45_total, uDrag);
fprintf('Drag Coefficient = %.4f ± %.4f\n', Cd45, uCd45);
fprintf('Reynolds Number = %.4e ± %.4e\n', Re45, uRe45);
%% Boundary Layer - 30 Hz
fprintf('\n');
fprintf('\n');
fprintf('Experiment 4: \n\n');
% rake
tap_r = 1.09 / 2; % mm
taps_mm = [1.40 2.88 4.03 5.28 6.40 7.96 8.98 10.50 11.72 13.07 19.42 ...
```

```
   25.60 31.91 38.55 44.83 51.06 57.62 63.83 70.12 76.44]; % distance from top
of tap to baseline in mm
taps = taps_mm - tap_r; % this is the distance from middle of tap to baseline
for taps 1-20
% data
hole1_lam = readmatrix('BL_Laminar');     % upstream laminar data
hole2_turb = readmatrix('BL_Turbulent');  % downstream turbulent data
% Remove columns 21-32
hole1_lam(:,21:32) = [];
hole2_turb(:,21:32) = [];
% Take mean across time samples
upstream = mean(hole1_lam);
downstream = mean(hole2_turb);
% Remove negative pressures
upstream(upstream < 0) = 0;
downstream(downstream < 0) = 0;
% Convert pressure to velocity
upstream_v = sqrt((2.*upstream)./0.00237)./0.3048;
downstream_v = sqrt((2.*downstream)./0.00237)./0.3048;
figure(6)
plot(upstream_v, taps, 'r-*', 'Linewidth', 1)
hold on
plot(downstream_v, taps, 'b-*', 'Linewidth', 1)
xlabel('Velocity (m/s)')
ylabel('Rake Location (mm)')
title('Boundary Layer 30Hz')
legend('Upsteam', 'Downstream', 'Location', 'best')
hold off
%% Airfoil: 30Hz and 45Hz
fprintf('\n');
fprintf('\n');
fprintf('Experiment 5: \n');
% Import Data
clean30 = readmatrix('M003_AF30Clean.txt');
dirty30 = readmatrix('M003_AF30Dirty.txt');
clean45 = readmatrix('M003_AF45Clean.txt');
dirty45 = readmatrix('M003_AF45Dirty.txt');
% Assign Columns
% 1 = AoA (deg),
% 2 = Normal Force (N),
% 3 = Side Force (N)
% 4 = Axial Force (N),
% 5 = Pitching Moment (N-m)
% 30 Hz
aoa_clean30 = clean30(:,1);
F_normal_clean30 = clean30(:,2);
F_side_clean30 = clean30(:,3);
F_axial_clean30 = clean30(:,4);
M_pitch_clean30 = clean30(:,5);
aoa_dirty30 = dirty30(:,1);
F_normal_dirty30 = dirty30(:,2);
F_side_dirty30 = dirty30(:,3);
F_axial_dirty30 = dirty30(:,4);
M_pitch_dirty30 = dirty30(:,5);
% 45 Hz
aoa_clean45 = clean45(:,1);
F_normal_clean45 = clean45(:,2);
```

```
F_side_clean45 = clean45(:,3);
F_axial_clean45 = clean45(:,4);
M_pitch_clean45 = clean45(:,5);
aoa_dirty45 = dirty45(:,1);
F_normal_dirty45 = dirty45(:,2);
F_side_dirty45 = dirty45(:,3);
F_axial_dirty45 = dirty45(:,4);
M_pitch_dirty45 = dirty45(:,5);
% Airfoil and Fluid Properties (SI)
rho = 1.225;                % kg/m^3
Chord = 0.08975;            % m
Span  = 0.255;              % m
WingArea = Chord * Span;    % m^2
V_30Hz = 27.727;            % m/s
V_45Hz = 43.269;            % m/s
% Compute Lift, Drag, Coefficients, L/D
% 30 Hz
Lift_clean30 = F_normal_clean30.*cosd(aoa_clean30) - F_axial_clean30.* ...
   sind(aoa_clean30);
Drag_clean30 = F_normal_clean30.*sind(aoa_clean30) + F_axial_clean30.* ...
   cosd(aoa_clean30);
CL_clean30 = Lift_clean30./(0.5*rho*V_30Hz^2*WingArea);
CD_clean30 = Drag_clean30./(0.5*rho*V_30Hz^2*WingArea);
CM_clean30 = M_pitch_clean30./(0.5*rho*V_30Hz^2*WingArea*Chord);
L_D_clean30 = Lift_clean30./Drag_clean30;
Lift_dirty30 = F_normal_dirty30.*cosd(aoa_dirty30) - F_axial_dirty30.* ...
   sind(aoa_dirty30);
Drag_dirty30 = F_normal_dirty30.*sind(aoa_dirty30) + F_axial_dirty30.* ...
   cosd(aoa_dirty30);
CL_dirty30 = Lift_dirty30./(0.5*rho*V_30Hz^2*WingArea);
CD_dirty30 = Drag_dirty30./(0.5*rho*V_30Hz^2*WingArea);
CM_dirty30 = M_pitch_dirty30./(0.5*rho*V_30Hz^2*WingArea*Chord);
L_D_dirty30 = Lift_dirty30./Drag_dirty30;
% 45 Hz
Lift_clean45 = F_normal_clean45.*cosd(aoa_clean45) - F_axial_clean45.* ...
   sind(aoa_clean45);
Drag_clean45 = F_normal_clean45.*sind(aoa_clean45) + F_axial_clean45.* ...
   cosd(aoa_clean45);
CL_clean45 = Lift_clean45./(0.5*rho*V_45Hz^2*WingArea);
CD_clean45 = Drag_clean45./(0.5*rho*V_45Hz^2*WingArea);
CM_clean45 = M_pitch_clean45./(0.5*rho*V_45Hz^2*WingArea*Chord);
L_D_clean45 = Lift_clean45./Drag_clean45;
Lift_dirty45 = F_normal_dirty45.*cosd(aoa_dirty45) - F_axial_dirty45.* ...
   sind(aoa_dirty45);
Drag_dirty45 = F_normal_dirty45.*sind(aoa_dirty45) + F_axial_dirty45.* ...
   cosd(aoa_dirty45);
CL_dirty45 = Lift_dirty45./(0.5*rho*V_45Hz^2*WingArea);
CD_dirty45 = Drag_dirty45./(0.5*rho*V_45Hz^2*WingArea);
CM_dirty45 = M_pitch_dirty45./(0.5*rho*V_45Hz^2*WingArea*Chord);
L_D_dirty45 = Lift_dirty45./Drag_dirty45;
% 30 Hz Plots
figure(7)
plot(aoa_clean30,Lift_clean30,'b-o',aoa_dirty30,Lift_dirty30,'r-s', ...
   'LineWidth',1.5)
title('30Hz: Angle of Attack vs Lift')
xlabel('Angle of Attack (deg)')
ylabel('Lift (N)')
```

```
legend('Clean','Dirty')
grid on
figure(8)
plot(aoa_clean30,Drag_clean30,'b-o',aoa_dirty30,Drag_dirty30,'r-s', ...
   'LineWidth',1.5)
title('30Hz: Angle of Attack vs Drag')
xlabel('Angle of Attack (deg)')
ylabel('Drag (N)')
legend('Clean','Dirty')
grid on
figure(9)
plot(aoa_clean30,M_pitch_clean30,'b-o',aoa_dirty30,M_pitch_dirty30, ...
   'r-s','LineWidth',1.5)
title('30Hz: Angle of Attack vs Pitching Moment')
xlabel('Angle of Attack (deg)')
ylabel('Pitching Moment (N-m)')
legend('Clean','Dirty')
grid on
figure(10)
plot(aoa_clean30,CL_clean30,'b-o',aoa_dirty30,CL_dirty30,'r-s', ...
   'LineWidth',1.5)
title('30Hz: Angle of Attack vs Coefficient of Lift')
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Lift')
legend('Clean','Dirty')
grid on
figure(11)
plot(aoa_clean30,CD_clean30,'b-o',aoa_dirty30,CD_dirty30,'r-s', ...
   'LineWidth',1.5)
title('30Hz: Angle of Attack vs Coefficient of Drag')
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Drag')
legend('Clean','Dirty')
grid on
figure(12)
plot(aoa_clean30,CM_clean30,'b-o',aoa_dirty30,CM_dirty30,'r-s', ...
   'LineWidth',1.5)
title('30Hz: Angle of Attack vs Coefficient of Moment')
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Moment')
legend('Clean','Dirty')
grid on
figure(13)
plot(aoa_clean30,L_D_clean30,'b-o',aoa_dirty30,L_D_dirty30,'r-s', ...
   'LineWidth',1.5)
title('30Hz: Angle of Attack vs Lift/Drag')
xlabel('Angle of Attack (deg)')
ylabel('Lift/Drag')
legend('Clean','Dirty')
grid on
% 45 Hz Plots
figure(14)
plot(aoa_clean45,Lift_clean45,'b-o',aoa_dirty45,Lift_dirty45,'r-s', ...
   'LineWidth',1.5)
title('45Hz: Angle of Attack vs Lift')
xlabel('Angle of Attack (deg)')
ylabel('Lift (N)')
```

```
legend('Clean','Dirty')
grid on
figure(15)
plot(aoa_clean45,Drag_clean45,'b-o',aoa_dirty45,Drag_dirty45,'r-s', ...
   'LineWidth',1.5)
title('45Hz: Angle of Attack vs Drag')
xlabel('Angle of Attack (deg)')
ylabel('Drag (N)')
legend('Clean','Dirty')
grid on
figure(16)
plot(aoa_clean45,M_pitch_clean45,'b-o',aoa_dirty45,M_pitch_dirty45, ...
   'r-s','LineWidth',1.5)
title('45Hz: Angle of Attack vs Pitching Moment')
xlabel('Angle of Attack (deg)')
ylabel('Pitching Moment (N-m)')
legend('Clean','Dirty')
grid on
figure(17)
plot(aoa_clean45,CL_clean45,'b-o',aoa_dirty45,CL_dirty45,'r-s', ...
   'LineWidth',1.5)
title('45Hz: Angle of Attack vs Coefficient of Lift')
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Lift')
legend('Clean','Dirty')
grid on
figure(18)
plot(aoa_clean45,CD_clean45,'b-o',aoa_dirty45,CD_dirty45,'r-s', ...
   'LineWidth',1.5)
title('45Hz: Angle of Attack vs Coefficient of Drag')
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Drag')
legend('Clean','Dirty')
grid on
figure(19)
plot(aoa_clean45,CM_clean45,'b-o',aoa_dirty45,CM_dirty45,'r-s', ...
   'LineWidth',1.5)
title('45Hz: Angle of Attack vs Coefficient of Moment')
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Moment')
legend('Clean','Dirty')
grid on
figure(20)
plot(aoa_clean45,L_D_clean45,'b-o',aoa_dirty45,L_D_dirty45,'r-s', ...
   'LineWidth',1.5)
title('45Hz: Angle of Attack vs Lift/Drag')
xlabel('Angle of Attack (deg)')
ylabel('Lift/Drag')
legend('Clean','Dirty')
grid on
% Final Combined Plot: AoA vs L/D for all 4 datasets
figure(21)
plot(aoa_clean30, L_D_clean30, 'b-o', aoa_dirty30, L_D_dirty30, 'r-s', ...
   aoa_clean45, L_D_clean45, 'c-o', 'LineWidth', 1.5);
hold on
plot(aoa_dirty45, L_D_dirty45, '-s', 'Color', [1 0.5 0], 'LineWidth', 1.5);  %
orange
```

```
hold off
title('Angle of Attack vs Lift/Drag for 30Hz and 45Hz Clean and Dirty')
xlabel('Angle of Attack (deg)')
ylabel('Lift/Drag')
legend('30Hz Clean','30Hz Dirty','45Hz Clean','45Hz Dirty')
grid on
% reynolds numbers
mu = 1.81e-5; % kg/(m·s)
Re_30Hz = rho * V_30Hz * Chord / mu
Re_45Hz = rho * V_45Hz * Chord / mu
% Peak efficiencies (30Hz, 45Hz)(clean, dirty)
peakEff_clean30 = max(L_D_clean30)
peakEff_dirty30 = max(L_D_dirty30)
peakEff_clean45 = max(L_D_clean45)
peakEff_dirty45 = max(L_D_dirty45)
% 30 Hz lift-curve slope
liftcurve30_slope = (CL_clean30(5) - CL_clean30(1)) / (aoa_clean30(5) ...
   - aoa_clean30(1)); % per deg
liftcurve30_rad   = liftcurve30_slope * (180/pi); % per rad
fprintf('30 Hz lift-curve slope: %.4f per deg, %.4f per rad\n', ...
   liftcurve30_slope, liftcurve30_rad);
% 45 Hz lift-curve slope
liftcurve45_slope = (CL_clean45(5) - CL_clean45(1)) / (aoa_clean45(5) ...
   - aoa_clean45(1)); % per deg
liftcurve45_rad   = liftcurve45_slope * (180/pi); % per rad
fprintf('45 Hz lift-curve slope: %.4f per deg, %.4f per rad\n', ...
   liftcurve45_slope, liftcurve45_rad);
%% 2D/3D Cylinders
fprintf('\n');
fprintf('\n');
fprintf('Experiment 6: \n\n');
% Import data
data = readmatrix('M003_2d3d.txt');
% Row 1: Cylinder 0 (Baseline)
% Row 2: Cylinder 1 (Longest, "2D")
% Row 3: Cylinder 2 (Medium)
% Row 4: Cylinder 3 (Shortest)
% Extract axial forces (drag forces because Aoa very small)
% Although Fd = Fn sin(α) + Fa * cos(α) would be slightly more accurate but
% axial force assumption will do
Af0 = data(1, 4);  % Baseline
Af1 = data(2, 4);  % Cylinder 1 long
Af2 = data(3, 4);  % Cylinder 2
Af3 = data(4, 4);  % Cylinder 3 short
% Cylinder diameters (caliper)
D1 = 48.17e-3;  % m (Cylinder 1)
D2 = 48.37e-3;  % m (Cylinder 2)
D3 = 47.98e-3;  % m (Cylinder 3)
% Cylinder lengths (ruler)
L1 = 0.607;  % m
L2 = 0.507;  % m
L3 = 0.267;  % m
% Fluid properties
rho = 1.225;  % kg/m^3
mu = 1.81e-5;  % kg/(m·s)
v30 = 27.727;  % m/s from calibration
% Measurement uncertainties
```

```
uD = 0.005e-3;    % diameter uncertainty (m)
uV = 0.00196;     % velocity uncertainty (m/s) also from calibration
% Reynolds numbers
Re0 = NaN;  % No cylinder, no characteristic length
Re1 = (rho * v30 * D1) / mu;
Re2 = (rho * v30 * D2) / mu;
Re3 = (rho * v30 * D3) / mu;
% Reynolds number uncertainties (velocity + diameter)
uRe0 = NaN;
uRe1 = Re1 * sqrt( (uV/v30)^2 + (uD/D1)^2 );
uRe2 = Re2 * sqrt( (uV/v30)^2 + (uD/D2)^2 );
uRe3 = Re3 * sqrt( (uV/v30)^2 + (uD/D3)^2 );
% front areas (Diameter × Length)
A1 = D1 * L1;
A2 = D2 * L2;
A3 = D3 * L3;
% Calculate drag (subtract baseline for cylinders, baseline is just Af0)
Drag0 = Af0;  % Baseline drag
Drag1 = Af1 - Af0;
Drag2 = Af2 - Af0;
Drag3 = Af3 - Af0;
% drag coefficients
pdyn = 0.5 * rho * v30^2;  % Dynamic pressure
CD0 = NaN;   % Undefined: no cylinder, no reference area
CD1 = Drag1 / (pdyn * A1);
CD2 = Drag2 / (pdyn * A2);
CD3 = Drag3 / (pdyn * A3);
% Uncertainties (±2% for force data)
uDrag0 = abs(Drag0) * 0.02;
uDrag1 = abs(Drag1) * 0.02;
uDrag2 = abs(Drag2) * 0.02;
uDrag3 = abs(Drag3) * 0.02;
uCD0 = NaN;
uCD1 = abs(CD1) * 0.02;
uCD2 = abs(CD2) * 0.02;
uCD3 = abs(CD3) * 0.02;
% Display results
fprintf('Cylinder 0 (Baseline - No Cylinder):\n');
fprintf('  Reynolds Number: %.4f ± %.4f\n', Re0, uRe0);
fprintf('  Drag = %.4e ± %.4e N\n', Drag0, uDrag0);
fprintf('  C_D = %.4f ± %.4f\n\n', CD0, uCD0);
fprintf('Cylinder 1 (Longest - "2D", L=%.3f m):\n', L1);
fprintf('  Reynolds Number: %.3e ± %.3e\n', Re1, uRe1);
fprintf('  Drag = %.4f ± %.4f N\n', Drag1, uDrag1);
fprintf('  C_D = %.4f ± %.4f\n\n', CD1, uCD1);
fprintf('Cylinder 2 (Medium, L=%.3f m):\n', L2);
fprintf('  Reynolds Number: %.3e ± %.3e\n', Re2, uRe2);
fprintf('  Drag = %.4f ± %.4f N\n', Drag2, uDrag2);
fprintf('  C_D = %.4f ± %.5f\n\n', CD2, uCD2);
fprintf('Cylinder 3 (Shortest, L=%.3f m):\n', L3);
fprintf('  Reynolds Number: %.3e ± %.3e\n', Re3, uRe3);
fprintf('  Drag = %.4f ± %.4f N\n', Drag3, uDrag3);
fprintf('  C_D = %.4f ± %.5f\n\n', CD3, uCD3);
% Create table for report (include Cylinder 0)
CylNum = [0; 1; 2; 3];
Lengths = [0; L1; L2; L3];
ReNumbers = [Re0; Re1; Re2; Re3];
```

```
uReNumbers = [uRe0; uRe1; uRe2; uRe3];
Drags = [Drag0; Drag1; Drag2; Drag3];
uDrags = [uDrag0; uDrag1; uDrag2; uDrag3];
CDs = [CD0; CD1; CD2; CD3];
uCDs = [uCD0; uCD1; uCD2; uCD3];
T = table(CylNum, Lengths, ReNumbers, uReNumbers, Drags, uDrags, CDs, ...
   uCDs, 'VariableNames', {'Cylinder', 'Length_m', 'Re', 'uRe', ...
   'Drag_N', 'uDrag_N', 'CD', 'uCD'});
disp(T);
```

6.5 AI Statement

My only use of AI was finishing touches on my lab report. Upon completion I copied my entire lab report into Claude followed by the prompt:

“Are there any spelling or grammatical errors in my lab report on fluid dynamics?”

In which Claude replied:

“

Fix these 3 errors:

Page 33: "flow rate in conserved" → "flow rate is conserved"
Page 10: "trailing vortices bring" → "trailing vortices form" (or complete the sentence)
Page 18: "Accelerometer just measures" → "The accelerometer just measures"

“

I fixed those three errors.